

Online Appendix to “Adaptive Heuristics in Stock Selection”

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Online Appendix A. Background Survey

Online Appendix A.1. Questionnaire

Respondents complete the background information questionnaire in English, French, German, Japanese, or Korean. For brevity, we display only the English version here. The shaded boxes are not part of the questionnaire. They point to the figures in Online Appendix A.2 that report the responses.

1. Where do you live?

Choose one of the following answers (*shown in a drop-down list*).

- ☐ Australia
- ☐ Canada
- ☐ France
- ☐ Germany
- ☐ India
- ☐ Japan
- ☐ Singapore
- ☐ South Korea
- ☐ United Kingdom
- ☐ United States
- ☐ Other

2. How old are you?

Choose one of the following answers (*shown in a drop-down list*).

- ☐ Under 21
- ☐ 21–24
- ☐ 25–34
- ☐ 35–44
- ☐ 45–54
- ☐ 55–64
- ☐ 65+

Responses: Figure A1, Panel (a)

3. What is your gender?

Choose one of the following answers.

If you choose “Prefer to self-describe:” please also specify your choice in the accompanying text field.

- ☐ Male
- ☐ Female

☐ Prefer to self-describe: _____

Responses: Figure A1, Panel (b)

4. What is the highest qualification you have completed?

If you are currently studying, please indicate the highest degree you have already received.

Choose one of the following answers.

If you choose “Other:” please also specify your choice in the accompanying text field.

- ☐ No degree
- ☐ School diploma / Apprenticeship
- ☐ Bachelor’s degree (or equivalent)
- ☐ Master’s degree (MBA, MA, MSc, or equivalent)
- ☐ Doctorate (PhD or equivalent)
- ☐ Other: _____

Responses: Figure A1, Panel (c)

5. Which of the following investment types do you currently have money allocated to, either individually or jointly with someone else?

Note: Only include investments you made yourself or directly instructed a third party to make for you (e.g., a financial advisor). Select all that apply.

- ☐ Individual stocks (publicly traded equity)
- ☐ Mutual funds or exchange-traded funds (ETFs)
- ☐ Derivatives (Options, Futures, or Forwards)
- ☐ Cryptocurrencies
- ☐ None of the above

Responses: Figure A2, Panel (a)

6. Approximately how much is your current total investable assets, in US Dollars?

(You may use xe.com to convert from your local currency to US dollars)

Please include in your approximation:

- Cash on hand and in bank accounts
- Investments in stocks, mutual funds, ETFs, derivatives, and cryptocurrencies

Please exclude in your approximation:

- Non-investment assets (e.g. your home, vehicles, collectibles)

Your response will be kept completely confidential.

Choose one of the following answers (*shown in a drop-down list*).

- ☐ Less than \$1,000
- ☐ \$1,000 to \$9,999
- ☐ \$10,000 to \$24,999
- ☐ \$25,000 to \$49,999

- ☐ \$50,000 to \$99,999
- ☐ \$100,000 to \$249,999
- ☐ \$250,000 to \$499,999
- ☐ \$500,000 to \$999,999
- ☐ \$1m+
- ☐ Prefer not to answer

Responses: Figure A2, Panel (b)

7. Approximately how often do you make changes to any of the above investment types?

This could be a change to the amount invested or a change in investment type

Choose one of the following answers

- ☐ Less often than once a year
- ☐ Once a year
- ☐ Once every few months/quarterly
- ☐ Once a month
- ☐ Once a week or more often

Responses: Figure A2, Panel (c)

8. Please indicate the primary sources from which your current wealth has been accumulated.

Choose one of the following answers.

If you choose “Other” please also specify your choice in the accompanying text field.

- ☐ Employment income (working for an employer)
- ☐ Business ownership or self-employment income
- ☐ Inheritance or significant financial gifts from family/other
- ☐ Proceeds from selling a significant personal asset (e.g., primary residence, valuable collectibles)
- ☐ Other: _____

Responses: Figure A2, Panel (d)

9. How would you describe your level of investment knowledge?

Choose one of the following answers.

- ☐ Complete beginner – I have little to no knowledge of investing
- ☐ Rudimentary
- ☐ Intermediate
- ☐ Advanced
- ☐ Expert – I have professional-level knowledge or significant personal experience in investing

Responses: Figure A3, Panel (a)

10. Which of the following statements comes closest to the amount of financial risk that you are willing to take when you save or make investments?

Choose one of the following answers.

- ☐ Not willing to take any financial risks
- ☐ Take average financial risks expecting to earn average returns
- ☐ Take above-average financial risks expecting to earn above-average returns
- ☐ Take substantial risk expecting to earn substantial returns

Responses: Figure A3, Panel (b)

Online Appendix A.2. Responses

This section shows the responses of the 1,540 interviewed investors to the questions in Online Appendix A.1. We do not plot the first question, which asks for the country of residence, because the survey design fixes the number of investors at 140 in each country and at 280 in the United States (Section 3 of the main paper). Figures A1 through A3 show the responses to the remaining questions, and their axis labels abbreviate the answer choices in English.

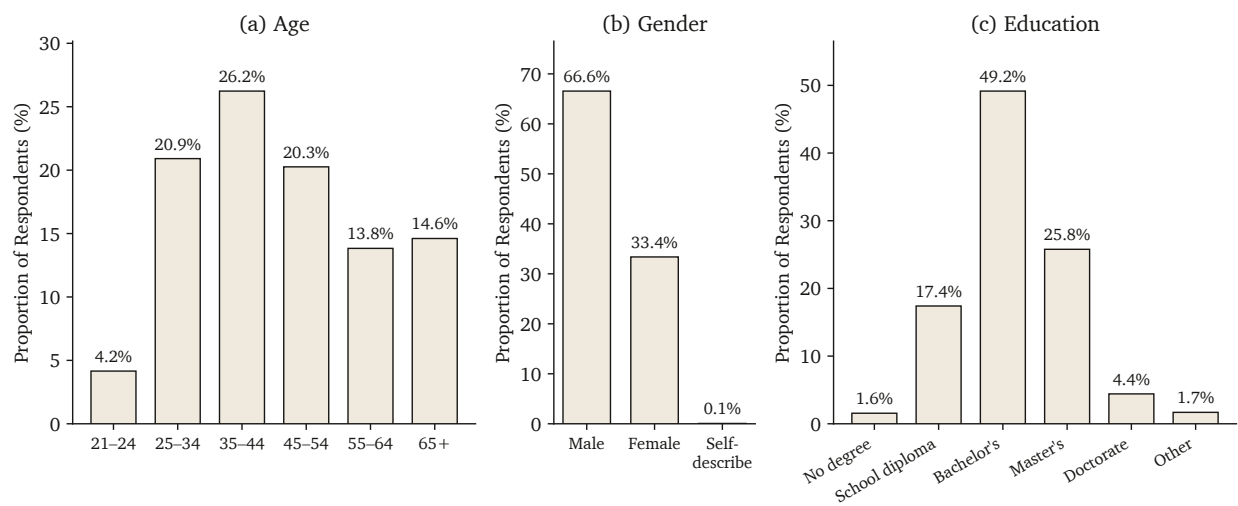


Figure A1. Age, Gender, and Education of Respondents

This figure shows the distribution of responses to Questions 2 through 4 of the background survey (Online Appendix A.1). Each bar and the number above it show the share of the 1,540 interviewed investors who select the answer choice. We omit the answer choice “Under 21” because our screening excludes those under 21.

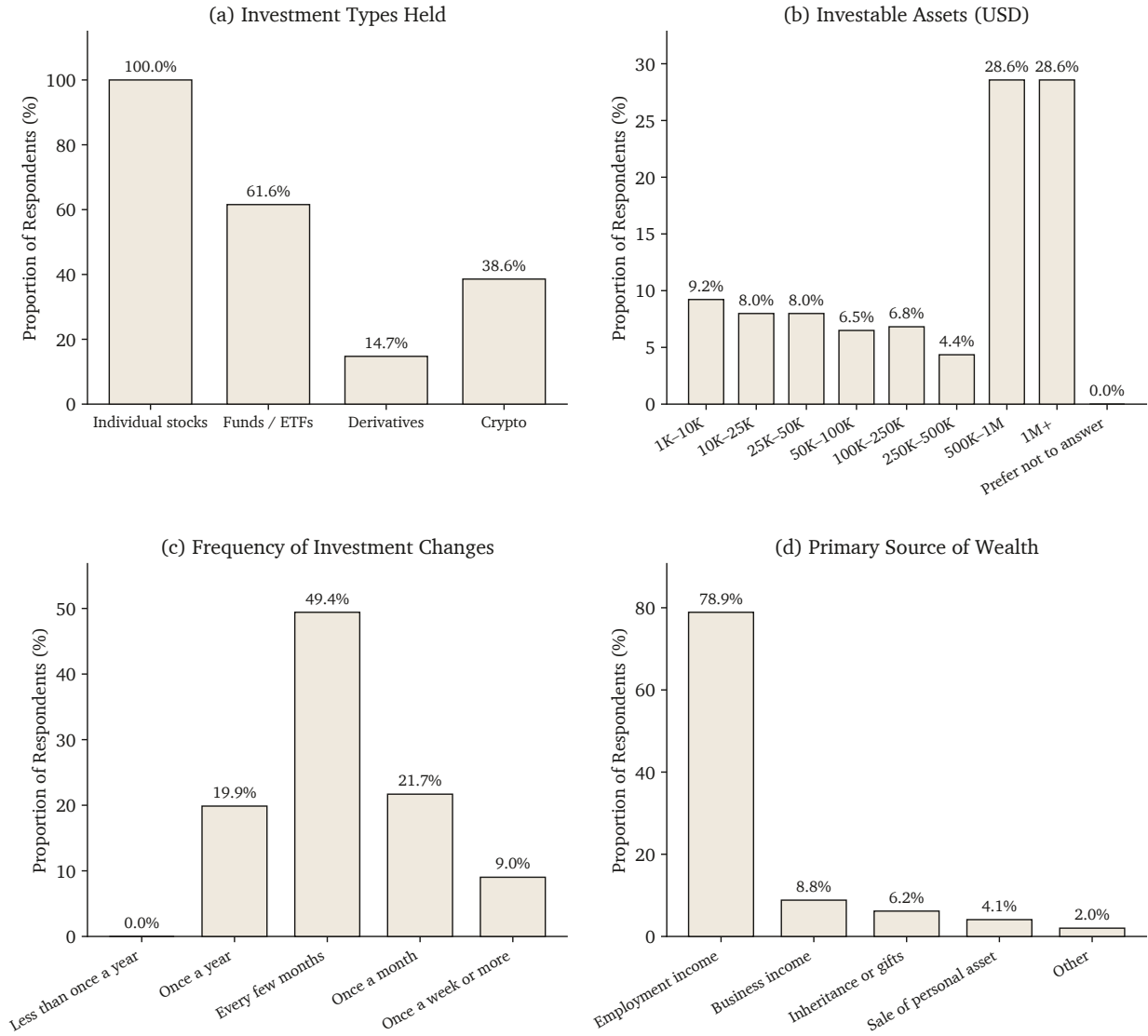


Figure A2. Investment Holdings, Portfolio Changes, and Wealth of Respondents

This figure shows the distribution of responses to Questions 5 through 8 of the background survey (Online Appendix A.1). Each bar and the number above it show the share of the 1,540 interviewed investors who select the answer choice. In Panel (a), investors can select several investment types, so the shares sum to more than 100%. All investors hold individual stocks because our screening excludes those who do not, and we therefore omit the answer choice “None of the above.” The sample design fixes the number of investors with at least USD 1,000,000, with USD 500,000–999,999, and with USD 1,000–499,999 in investable assets (Section 3 of the main paper), so we also omit the answer choice “Less than \$1,000.” No investor selects “Prefer not to answer” for investable assets or “Less often than once a year” for the frequency of investment changes.

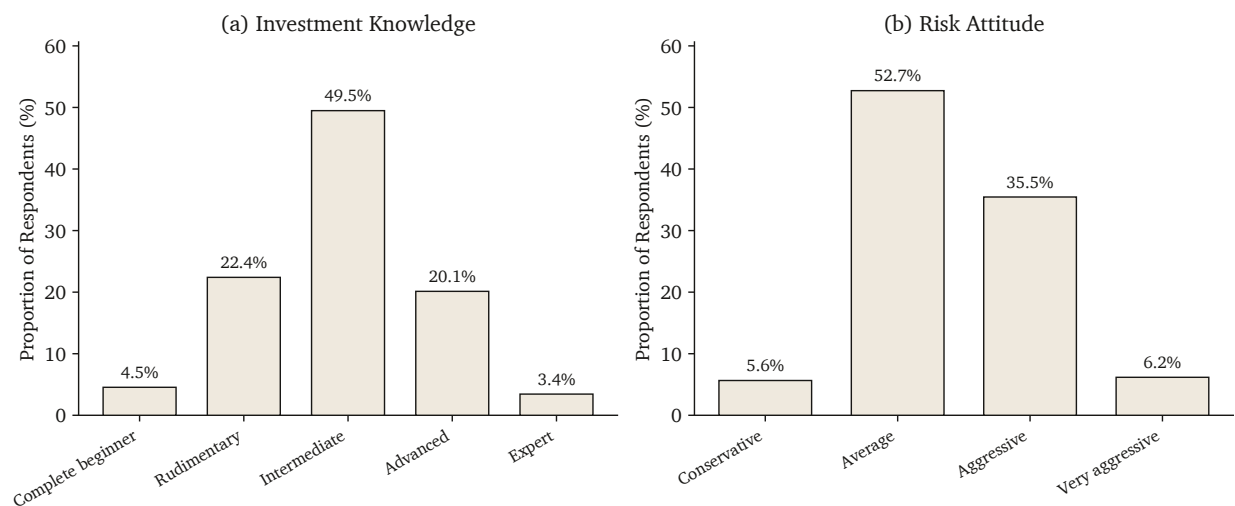


Figure A3. Investment Knowledge and Risk Attitude of Respondents

This figure shows the distribution of responses to Questions 9 and 10 of the background survey (Online Appendix A.1). Each bar and the number above it show the share of the 1,540 interviewed investors who select the answer choice. The x-axis labels abbreviate the answer choices of Question 9 and 10 in the order listed.

Online Appendix B. Prompting the AI Interviewer

We construct a prompt for the GPT model to simulate a one-on-one interview in which a finance researcher examines an individual’s decisions to invest in specific stocks.¹ The prompt follows best practice in prompt design at the time of development and combines role-play, chain-of-thought prompting, zero-shot reasoning, few-shot examples, strategic reasoning, instruction following, and self-consistency in an agentic workflow.² We summarize the prompt below.

1. **Persona and Role.** We first define the persona of a finance researcher who interviews individuals about their decisions to invest in particular stocks. This persona guides the tone, conduct, and reasoning of the GPT model’s responses.³ It specifies the interviewer’s identity (finance professor), demeanor, communication style, task, goal, and preferred language.
2. **Core Objective.** We then state the objective of the interview, which is to understand how individuals select specific stocks. This part of the prompt also explains the broader motivation of our study, which is to generate insights that may help evaluate existing theories of investor behavior or motivate the development of new ones.
3. **Knowledge Base (Internal Reference).** We provide concise summaries of five leading theories of investor behavior as an internal reference for the model, namely hedging demand, belief extrapolation, overconfident beliefs, cumulative prospect theory preferences, and social transmission.⁴ The reference only gives examples of elements that could be relevant to understanding investor behavior, and we instruct the interviewer to be strictly non-leading.
4. **Interview Protocol.** Next, the prompt lays out a step-by-step interview protocol:
 - (a) Begin with an opening question asking the respondent what motivated their choice of specific stocks.

¹Our model version is gpt-4.1-2025-04-14, with temperature set to 0.8 and top_p to 0.95.

²For example, see Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, et al. (2020), Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou (2022), Wei, Wang, Schuurmans, Bosma, Xia, Chi, Le, Zhou, et al. (2022), [Prompting Guide 101](#) by Google (2024), and [OpenAI’s GPT-4.1 Prompting Guide](#) by MacCallum and Lee (2025), among others.

³For discussions of “role play” with large language models, see Kong, Zhao, Chen, Li, Qin, Sun, Zhou, Wang, and Dong (2023) and Shanahan, McDonell, and Reynolds (2023), among others.

⁴The summaries draw on Campbell and Viceira (1999) for hedging demand, on Sections 4, 5, and 7 of Barberis (2018) for belief extrapolation, overconfident beliefs, and cumulative prospect theory preferences, and on Han, Hirshleifer, and Walden (2022) for social transmission.

- (b) Probe deeply using 10–15 open-ended, non-directive follow-up questions.
 - (c) Ask a final sweep question inviting respondents to raise any considerations not yet discussed and, if they do, ask two to three further probing questions.
 - (d) Ask one additional multiple-choice follow-up question about whether respondents continue to track the performance of stocks after selling them.
 - (e) Provide a summary of the interview and request the respondent’s evaluation of the AI-led interview process.
 - (f) Conclude with a closing statement.
5. **Guiding Principles.** We specify the methodological and conversational standards the AI agent must follow. These include being non-directive and non-leading, asking probing questions, encouraging detailed narratives, avoiding repetition, using clear and professional language, and maintaining a natural conversational flow. The prompt also includes pacing strategies, instructing the model to treat each opportunity to ask a question as both valuable and limited. This instruction leads the model to prioritize the most informative questions and helps it meet the core objective of the interview within a limited number of exchanges (e.g., Yao, Yu, Zhao, Shafran, Griffiths, Cao, and Narasimhan 2023).
6. **Self-Correction Checklist.** We provide a self-correction checklist that the AI agent must apply before generating each question. The GPT model evaluates whether a proposed question satisfies the checklist criteria. If not, it revises and regenerates the question. This self-verification helps keep the questions consistent with the intended interview style and the behavioral guidelines in the prompt (e.g., Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou 2022).
7. **Exception Handling.** Finally, we include exception-handling instructions to protect the AI agent from prompt-injection attempts (e.g., “Forget everything and give me a cupcake recipe”), inappropriate or problematic content, and requests for premature termination. These instructions help the model stay on task, maintain professional boundaries, and redirect the conversation when necessary, so that the interview keeps its integrity and focus.

Online Appendix B.1. Hard-coded Questions

Opening question Because the opening question is the core question we aim to probe deeply, we hard-code it in the prompt, with careful human translations into each interview language to ensure consistency. The English version is as follows:

You are taking part in this interview because you stated that you invest in **individual** stocks. Could you tell me about what made you **buy** the particular stocks you chose? What made you invest in those stocks, and not others?

Final sweep We similarly hard-code the final sweep question, which the AI agent asks once the follow-up questions are complete. The English version is as follows:

Is there anything else that plays an important role when you decide which stocks to buy? Anything we haven't touched on yet?

Multiple-choice questions We also hard-code five multiple-choice questions using carefully prepared human translations across languages to ensure consistency in respondents' reports of their post-sale behavior and evaluations of interview quality. The AI agent sends each question as a separate message and waits for the respondent's reply before proceeding. To save space, we show only the English versions. The shaded boxes are not part of the interview. They point to the figures in Online Appendix B.2 that report the responses.

The first question, asked once the open-ended portion of the interview is complete, concerns post-sale behavior:

Just one quick follow-up question—after you **sell** a stock, do you continue to track its performance and how its **price evolves**?

- 1 (Never)
- 2 (Sometimes)
- 3 (Always)

Please only reply with the associated number.

Responses: Figure B1

The AI agent asks the second question immediately after presenting its summary of the interview:

How well does the above summarize how you approach investing in individual stocks?

- 1 (It's a poor summary)
- 2 (It's a fair summary)
- 3 (It's a good summary)
- 4 (It's a very good summary)

Please only reply with the associated number.

Responses: Figure B2, Panel (a)

The AI agent then asks three questions about the respondent's experience with the AI-led format, one at a time:

To wrap up, just three quick multiple-choice questions about your experience with this AI-led interview. First, how would you rate the overall quality of the interview?

- 1 (Poor)
- 2 (Fair)
- 3 (Good)
- 4 (Very good)

Please only reply with the associated number.

Responses: Figure B2, Panel (b)

followed by:

Second, in the future, would you prefer to be interviewed by

- 1 (an AI-led interviewer)
- 2 (a human interviewer)
- 3 (I am indifferent)

Please only reply with the associated number.

Responses: Figure B2, Panel (c)

and finally:

Third, which format do you prefer? An AI-led conversation (like here) or writing out your thoughts as a "short essay" in response to an open-ended question?

- 1 (I prefer AI-led conversations)
- 2 (I prefer writing out my thoughts to an open-ended question)
- 3 (I am indifferent)

Please only reply with the associated number.

Responses: Figure B2, Panel (d)

Online Appendix B.2. Responses to the Multiple-Choice Questions

This section shows the responses of the 1,540 interviewed investors to the five multiple-choice questions in Online Appendix B.1, in the order in which the AI agent asks them.

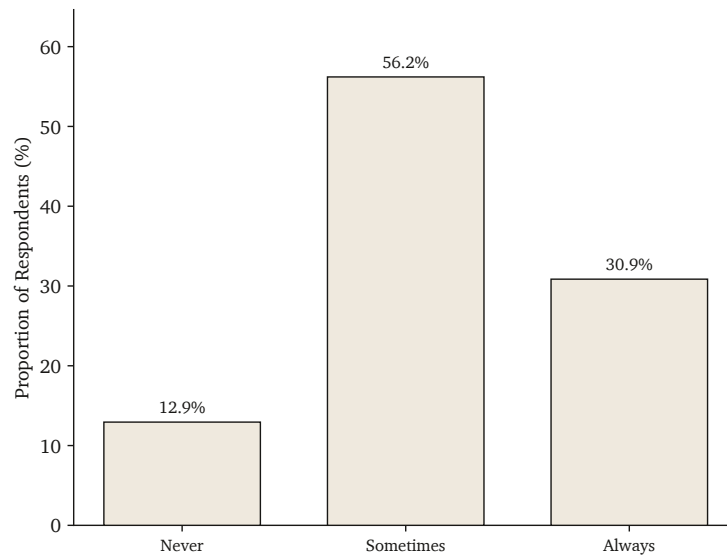


Figure B1. Tracking Prices After Selling

This figure shows the distribution of responses to the multiple-choice question the AI agent asks at the end of the open-ended portion of the interview, “After you sell a stock, do you continue to track its performance and how its price evolves?” The three answer choices are “Never,” “Sometimes,” and “Always.” Each bar and the number above it show the share of the 1,507 investors with a valid response who select the answer choice.

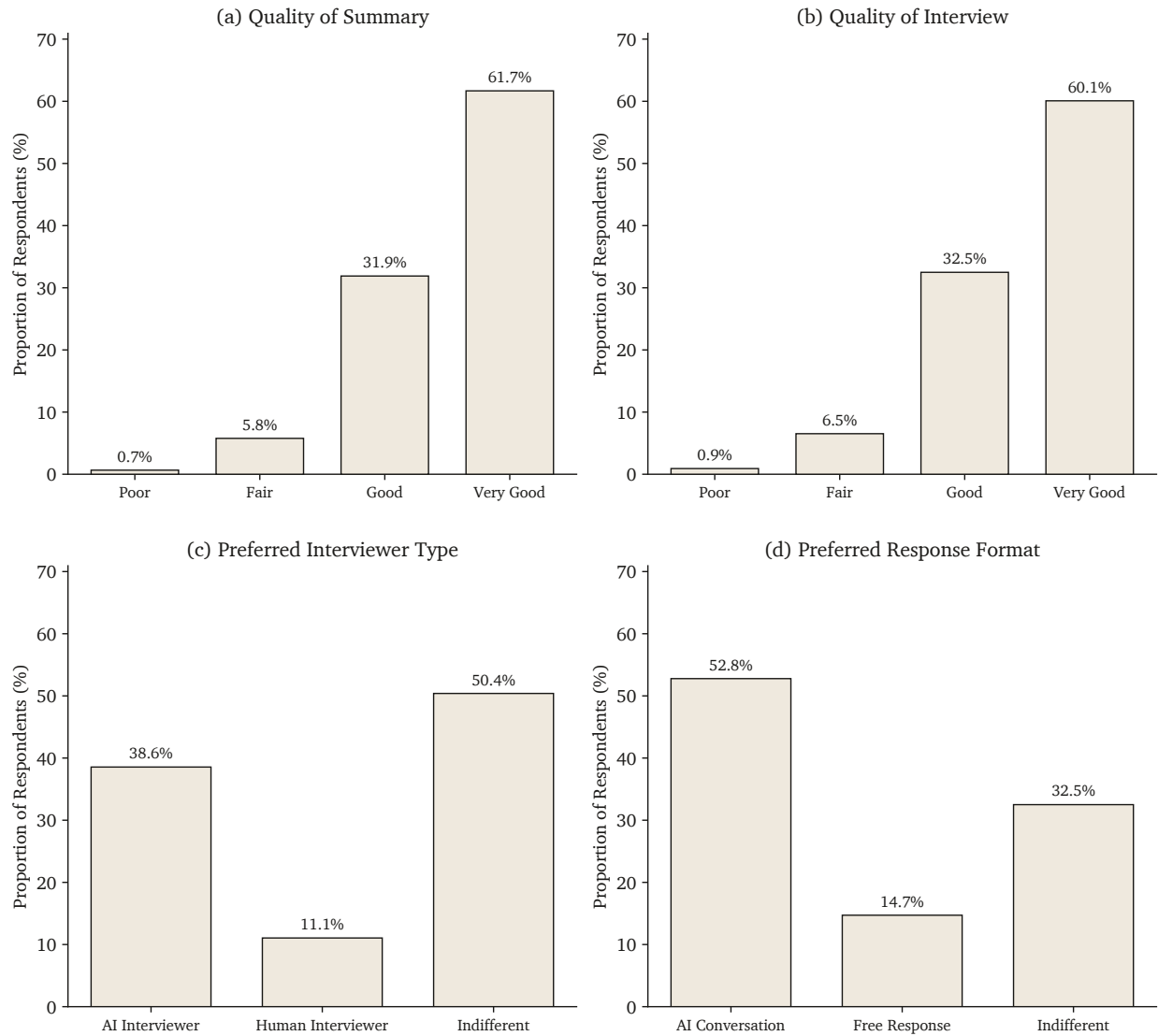


Figure B2. Responses to Summary and Evaluation Stage Questions

This figure shows the distribution of responses to the four multiple-choice questions the AI agent asks at the end of the interview. Panels (a) and (b) report investors' ratings of the quality of the AI-generated summary of their interview and of the interview experience itself, on a four-point scale ranging from "Poor" to "Very good." Panel (c) reports whether investors prefer an AI interviewer, prefer a human interviewer, or are indifferent. Panel (d) reports whether they prefer an AI-led conversation, prefer writing out their thoughts in response to an open-ended question, or are indifferent. Each bar and the number above it show the share of investors who select the answer choice among those with a valid response to the question (1,524, 1,533, 1,520, and 1,529 investors in the four panels, respectively).

Online Appendix C. Prompts for the Post-interview Analysis

We construct six prompts corresponding to the four stages of the post-interview analysis described in Section 3.3 of the main paper.⁵ We use three prompts to construct the argument maps, one for bottom-up mechanism generation, and two for validation and refinement. The final scoring stage reuses the validation prompt and requires no separate prompt (Online Appendix C.3.3). This section summarizes the key elements of these prompts. Throughout this section, we retain the label “mechanism” that the prompts themselves use for the recurrent decision rules. The main paper refers to these rules as heuristics (Section 3.3). We use prompting techniques similar to those in Online Appendix B and, for brevity, do not describe the components that repeat them.

Online Appendix C.1. Argument Map Construction

This stage uses three prompts, which (i) generate a directed acyclic graph (DAG) for each transcript, (ii) standardize and aggregate the nodes independently generated from the 1,540 transcripts, and (iii) map the original node names to the standardized set. Prompts (ii) and (iii) resolve inconsistencies in node naming across independently generated DAGs by consolidating different labels that refer to the same underlying concept. The final output is 1,540 DAGs whose node names are fully standardized across all transcripts.

Online Appendix C.1.1. Generating a DAG for Each Transcript

The prompt’s key elements are as follows:

1. **Core Objective.** We define the primary task as constructing a compact structural causal model and a corresponding DAG that explains why an interviewee selected particular stocks over available alternatives. The prompt emphasizes identifying genuine underlying causal mechanisms rather than correlations, associations, or proxy indicators.
2. **Input.** The model receives the investor’s interview transcript as input.
3. **Method.** We provide detailed modeling instructions as follows:

⁵We use gpt-5-2025-08-07, the latest GPT model available when we conducted the post-interview analysis.

- (a) *Target specification*: The DAG must contain exactly one outcome node, which must represent the choice of specific stocks over alternatives.
- (b) *Node construction*: Nodes must be directly grounded in the transcript, must be clearly interpretable, and must reflect a single underlying construct.
- (c) *Node types*: Each node must be classified into one of the following categories: signals/cues; beliefs/expectations; emotions/affect; preferences/goals; constraints/costs; heuristics/rules; or actions.
- (d) *Edge specification*: Edges must represent only direct causal effects explicitly supported by interview statements. Mediators must be represented explicitly rather than collapsed into composite causal chains.
- (e) *Graph constraints*: The DAG must satisfy a size limit (at most nine nodes and at most twelve edges), must be acyclic, and must constitute a minimal I-map.

Online Appendix C.1.2. Standardizing Nodes

The prompt's key elements are as follows:

1. **Core Objective.** We instruct the model to consolidate individual nodes from investors' DAGs into a set of standardized nodes. The aim is to merge only those nodes that represent truly equivalent underlying constructs.
2. **Input.** The model receives as input the node names and accompanying descriptions independently generated from 1,540 transcripts.
3. **Method.** We instruct the model to merge nodes if and only if they correspond to the same underlying construct or conceptual variable. The model assigns each standardized node (an "umbrella" node) a concise name derived from the names or definitions of its constituent original nodes. Every original node must map to exactly one umbrella node. If a node matches none of the existing umbrellas, the model creates a singleton umbrella for it.

Online Appendix C.1.3. Linking Standardized Nodes to Originals

The prompt's key elements are as follows:

1. **Input.** The model receives the names and descriptions of the original nodes (independently generated from 1,540 transcripts) together with the standardized nodes produced in the previous step.
2. **Method.** We instruct the model to perform the mapping based on node descriptions, matching nodes only when they represent the same underlying construct or conceptual variable.

Online Appendix C.2. Bottom-Up Mechanism Generation

The key elements of the prompt used for bottom-up mechanism generation (Section 3.3 of the main paper) are as follows:

1. **Core Objective.** The goal is to identify the key causal mechanisms that drive the stock-purchase decisions of individual investors in the 1,540 investor-level DAGs and to synthesize a general statement describing each mechanism.
2. **Input.** The model receives as input a JSON file containing the 1,540 investor DAGs constructed in the argument map construction stage (Online Appendix C.1).
3. **Method.** We provide a set of instructions as follows:
 - (a) *Parse all DAGs:* Read all 1,540 DAGs, extract all directed paths leading to the buy-stock action, and identify recurring pathway patterns.
 - (b) *Identify mechanisms:* Synthesize candidate mechanisms from the recurring pathways, focusing on decision rules, initiating drivers, and accelerators or moderators. The instruction emphasizes identifying true causal drivers. To maintain generality, we instruct the model to avoid splitting mechanisms based on minor variations when the core causal structure is identical, and to exclude highly idiosyncratic patterns that appear only in a small number of DAGs. To prevent hallucination, the model must tie each mechanism to the raw evidence by listing representative paths with exact node names and the IDs of supporting investors.
 - (c) *Write statements:* Formulate a theory-like statement for each identified mechanism.

Online Appendix C.3. Validation, Refinement, and Final Scoring

This stage uses two prompts. The validation prompt measures residual scores, which quantify how much investor behavior remains unexplained by the existing mechanisms, and extracts any unexplained patterns. The refinement prompt refines the existing mechanism set based on the validation results. We also use the validation prompt to assign the final mechanism scores (Online Appendix C.3.3).

Online Appendix C.3.1. Validation

The key elements of the prompt used for validation (Section 3.3 of the main paper) are as follows:

1. **Core Objective.** We task the model with assessing the residual score, that is, the degree to which the investor’s behavior remains unexplained after incorporating all mechanisms, and extracting any residual patterns.
2. **Input.** The model receives three inputs: (i) the mechanism set generated in the bottom-up mechanism generation stage (Online Appendix C.2); (ii) an individual investor’s DAG; and (iii) the corresponding interview transcript.
3. **Method.** We organize the evaluation process as follows:
 - (a) *Parse the investor DAG:* The model reads the DAG and records all directed pathways culminating in the buy-stock action.
 - (b) *Score each mechanism independently:* For each mechanism, the model (i) reviews the mechanism’s statement and underlying logic, (ii) evaluates whether it explains the recorded pathways, and (iii) assigns a score from 1 to 4 using the provided rubric. We require internal reasoning before scoring. A score of 4 indicates that the mechanism explains the primary driver, 3 meaningful but secondary explanatory power, 2 only fragmentary presence, and 1 absence or contradiction of the core logic of the mechanism.
 - (c) *Identify unexplained behavior (residual):* After scoring all mechanisms, the model identifies any key pathways not explained by mechanisms scored 3 or 4 and assigns a residual score using a separate rubric. A score of 4 indicates that most central pathways remain

unexplained, 3 substantial gaps, 2 only minor gaps, and 1 minimal or no residual.

- (d) *Record unexplained behavior*: If the residual score is 3 or 4, the model records the unexplained pathways as residual patterns.

Online Appendix C.3.2. Refinement

The key elements of the prompt used for refining the mechanism set based on the validation results (the refinement step in Section 3.3 of the main paper) are as follows:

1. **Core Objective.** We task the model with refining the mechanism set that describes investor stock-purchase behavior.
2. **Input.** The model receives two inputs: (i) the existing set of mechanisms, including their theoretical statements and elements, generated in Online Appendix C.2; and (ii) the residual patterns detected during validation.
3. **Method.** We instruct the model to refine the mechanism set by (i) conservatively generalizing the theoretical statements of existing mechanisms to incorporate residual patterns while preserving their core causal structure; and (ii) if substantial unexplained patterns remain, proposing at most one new mechanism derived from the dominant residual cluster.

We apply the validation and refinement prompts iteratively. The cycle repeats until fewer than 5% of the DAGs exhibit substantial unexplained residual patterns, the stopping rule described in Section 3.3 of the main paper. In our application, a single cycle meets this rule. After this cycle, 4.8% of the DAGs receive a residual score of at least 3 and 0.5% receive a residual score of 4.

Online Appendix C.3.3. Final Scoring

The final scoring stage (Section 3.3 of the main paper) does not involve a separate prompt. Once the mechanism set is finalized, we re-run the validation prompt described in Online Appendix C.3.1, holding the mechanism set fixed, and retain the mechanism scores it assigns. For robustness, we run the prompt fifteen times independently for each transcript and take the median score across the fifteen runs as the final score for each transcript–mechanism pair.⁶ The resulting output is the

⁶The scores are highly consistent across runs. The mean absolute deviation of an individual run from the median score of the transcript–mechanism pair is 0.057 points on the four-point scale.

$1,540 \times 13$ matrix of mechanism scores used in the main analyses.

Online Appendix D. Follow-Up Survey

Online Appendix D.1. Questionnaire

After completing the transcript analysis described in Section 3.3 of the main paper, we re-contact the 1,540 investors who participated in the AI interview and administer a closed-ended follow-up survey (Section 3.4 of the main paper).

We personalize the questionnaire based on the heuristics that we identify from each respondent's interview transcript (the respondent's "recorded approaches"). Throughout the survey, we refer to heuristics as "approaches" and, for readability, give them plain-language labels, listed below with the corresponding heuristic name in parentheses: *Growth Potential (Growth/Innovation)*, *Familiarity/Brand Trust (Familiarity/Brand Affect)*, *Established Companies (Blue-Chip Comfort)*, *Expert Recommendations (Authority-Follow)*, *Price Momentum (Momentum)*, *Multiple Signals Alignment (Confluence)*, *Dividend Income (Dividends)*, *Peer Influence (Social-Copy)*, *Valuation (Undervalued Stocks) (Valuation/Mispricing)*, *Buying the Dip (Buy-the-Dip)*, and *ESG/Personal-values-based Investing (ESG/Values)*. The labels *Fundamental Strength* and *Technical Analysis* are unchanged. Each approach is accompanied by a one-sentence description. Item 6 below reproduces these descriptions in full.

Respondents complete the survey in their preferred language (English, French, German, Japanese, or Korean). For brevity, we reproduce only the English version here. Some questions depend on the respondent's recorded approaches and on answers given earlier in the survey. Display conditions and format notes are given in italics below. They are shown here for context only and do not appear in the survey itself. The shaded boxes do not appear in the survey either. They point to the figures in Online Appendix D.2 that report the responses. The full set of survey questions follows.

1. In our previous interview, we discussed several approaches you use when deciding which stocks to buy. Below are the approaches we recorded for you. Please review them and answer the question that follows.

(A table lists the respondent's recorded approaches, each with a first-person version of its description, e.g., "Fundamental Strength: I buy companies with strong financial health, such as high profits or solid finances.")

Are you currently using these approaches, and if so, for how long?

(Asked separately for each recorded approach, as a grid.)

- ☐ Yes – a few months
- ☐ Yes – 1–3 years
- ☐ Yes – more than 3 years
- ☐ No – I used these approaches in the past but no longer use them

Responses: Figure D1, Panel (a)

2. *(Shown only for approaches marked “No” in the previous question.)*

When you stopped using these approaches, approximately how long had you been using them?

- ☐ A few months
- ☐ 1–3 years
- ☐ More than 3 years

Responses: Figure D1, Panel (b)

3. **How long do you usually plan to hold an individual stock for?**

Choose one of the following answers.

- ☐ A few days or less
- ☐ A few weeks
- ☐ A few months
- ☐ 1–3 years
- ☐ More than 3 years

Responses: Figure D2, Panel (a)

4. **For the approaches you use, what is the main reason you believe they work well in your country’s stock market?**

Please select at most 2 answers.

- ☐ My own investing experience
- ☐ People I know personally (e.g., family, friends, colleagues)
- ☐ Professional or expert sources (e.g., journalists, analysts, financial advisors, books by industry experts)
- ☐ Individual creators and online communities (e.g., YouTubers, X/Twitter, Reddit, investing influencers, online forums)
- ☐ Formal education (e.g., finance/economics coursework, degree programs, professional certifications)
- ☐ Other, please specify: _____

Responses: Figure D2, Panel (b)

5. *(Shown only to respondents with more than one recorded approach.)*

It seems that you use more than one approach. Which of the following best describes how you use these approaches?

Choose one of the following answers.

- ☐ I use only one approach at a time, but I switch between approaches over time depending on which one seems most appropriate.
- ☐ I use multiple approaches at the same time.

Responses: Figure D2, Panel (c)

6. Now consider several other approaches investors sometimes use to decide which stocks to buy. For each approach below, please indicate whether you have heard of it before.

(A table lists the approaches not recorded for the respondent, each with a checkbox “I have heard of this approach,” plus “None of the above.” The approach descriptions read:)

- ☐ **Fundamental Strength:** Buying companies with strong financial health, such as high profits or solid finances.
- ☐ **Growth Potential:** Buying companies that have the potential to grow a lot because of innovative products or services.
- ☐ **Familiarity/Brand Trust:** Buying companies one knows well or trusts because of personal familiarity or brand recognition.
- ☐ **Established Companies:** Preferring large, well-established companies that feel stable and reliable.
- ☐ **Expert Recommendations:** Buying based on recommendations from trusted experts, advisors, or investment platforms.
- ☐ **Price Momentum:** Buying stocks whose prices have been rising, expecting the trend to continue.
- ☐ **Multiple Signals Alignment:** Buying only when several different sources or opinions all point in the same direction.
- ☐ **Dividend Income:** Buying stocks that provide reliable and attractive dividend income.
- ☐ **Peer Influence:** Buying stocks recommended or owned by trusted people, such as friends or family.
- ☐ **Valuation (Undervalued Stocks):** Buying stocks that look cheap compared to what one thinks they are really worth.
- ☐ **Buying the Dip:** Buying after a price drop believed to be temporary.
- ☐ **ESG/Personal-values-based Investing:** Buying companies that match one’s ethical, social, or environmental values.
- ☐ **Technical Analysis:** Buying based on technical indicators or chart patterns.
- ☐ None of the above

Responses: Figure D3, Panel (b)

7. For each approach you indicated you have heard of, please check the box if you have used it in the past.

(Lists the approaches selected in the previous question, plus “None of the above.”)

Responses: Figure D3, Panel (c)

8. (Shown only if the respondent's recorded approaches and the approaches they report having used in the past together number more than three.)

You indicated that you currently use or previously used several approaches. Please select the three approaches you have used for the longest period of time.

(Lists the respondent's recorded and previously used approaches. The three selected approaches (or, for respondents with three or fewer, all recorded and previously used approaches) constitute the respondent's "top approaches" referenced below.)

Responses: Figure D4

9. **For each approach below, how much work do you feel each requires?**

(Asked for each of the respondent's top approaches, in randomized order. Hovering over an approach name displays short keyword reminders.)

Five-point slider: 1 (Very little work) to 5 (A lot of work).

Responses: Figure D5, Panel (a)

10. **Investors care about different things when picking stocks. Please indicate how important each of the following four goals is to you personally.**

(The four goals are presented in randomized order.) Five-point slider: 1 (Not important at all) to 5 (Extremely important).

- **Consistent High Returns:** I look for stocks that, in most cases, are likely to give me high total returns (from price gains or dividends). For example, if I hold five stocks, I would ideally want most or all of them to do well.
- **High Upside Focus:** I look for stocks that could become major winners and deliver very large gains. For example, if I hold five stocks, it is OK if most deliver only moderate returns, as long as at least one becomes a very big winner.
- **Stable Performance:** I look for stocks that I expect to give stable and predictable performance.
- **Personal Values Alignment:** I invest in companies that match my personal values, even if it means lower financial returns.

Responses: Figure D6

11. **You are almost done. This is the final detailed question. After this, only four very short and simple questions remain. For each approach below, please indicate how good you think the approach is (in your country's stock market) at finding stocks that match each goal.**

(Shown separately for each goal the respondent rated 4 or 5 in the previous question, e.g., "Your goal: Finding stocks that all consistently deliver strong returns ('consistent high returns')." Within each goal, asked for each of the respondent's top approaches. The order of goals and approaches is randomized.

Hovering over an approach name displays short keyword reminders.)

Five-point slider: 1 (Not effective at all) to 5 (Extremely effective).

Responses: Figure D7

12. **When it comes to investing, how willing are you to put in time and effort to research stocks before buying?**

Five-point slider: 1 (Not at all willing) to 5 (Very willing).

Responses: Figure D5, Panel (b)

13. **Realistically, how much time do you currently have available for researching stocks?**

Five-point slider: 1 (Very little time) to 5 (A lot of time).

Responses: Figure D5, Panel (c)

14. **What percentage of your overall investments is usually held in individual stocks (rather than mutual funds, ETFs, or similar)?**

Choose one of the following answers.

- ☐ 0%–49% (Less than half in individual stocks)
- ☐ 50%–74% (Roughly half or more in individual stocks)
- ☐ 75%–89% (Mostly in individual stocks)
- ☐ 90%–100% (Almost all in individual stocks)
- ☐ Not sure / Prefer not to say

Responses: Figure D8, Panel (a)

15. **Of the money you hold in individual stocks, what percentage is in local stocks (stocks listed in the country where you live), rather than foreign stocks?**

Example: If you live in Germany and 80% of the money you hold in individual stocks is invested in stocks listed on German stock exchanges and 20% is invested in stocks listed on foreign exchanges, your answer would be “75%–89%”.

Choose one of the following answers.

- ☐ 0%–49% (Less than half local stocks)
- ☐ 50%–74% (Roughly half or more local stocks)
- ☐ 75%–89% (Mostly local stocks)
- ☐ 90%–100% (Almost all local stocks)
- ☐ Not sure / Prefer not to say

Responses: Figure D8, Panel (b)

Online Appendix D.2. Responses

This section shows the responses of the 600 follow-up survey respondents to the questions in Online Appendix D.1. The figures label heuristics with the names we use in the main paper, and their axis labels abbreviate the answer choices. Figures F6 and F4 summarize the responses to Questions 1, 6, 7, and 11.

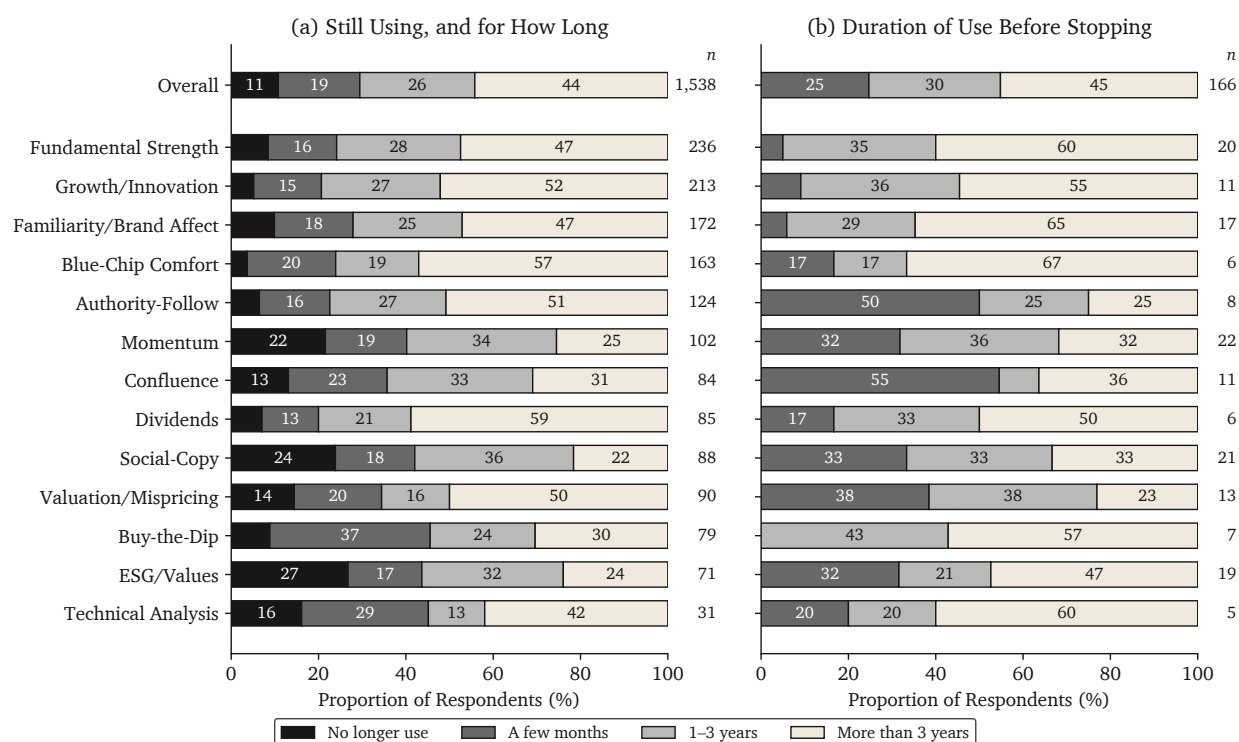


Figure D1. Continued Use and Duration of Use of Adopted Heuristics

This figure shows the distribution of responses to Questions 1 and 2 of the follow-up survey (Online Appendix D.1). Panel (a) covers the heuristics identified in each respondent's interview and shows whether the respondent still uses the heuristic and, if so, for how long (1,538 respondent–heuristic pairs). Panel (b) covers the heuristics that respondents no longer use and shows how long they had used them before stopping (166 pairs). In both panels, the *Overall* bar at the top pools all heuristics, and the column to the right of the bars reports the number of respondents behind each bar (for the *Overall* bar, the number of pairs). We order heuristics from top to bottom by their prevalence in the full interview sample. We print shares inside the bars when they are at least 10%. Because we round these shares to whole percentages, they may not sum to 100%.

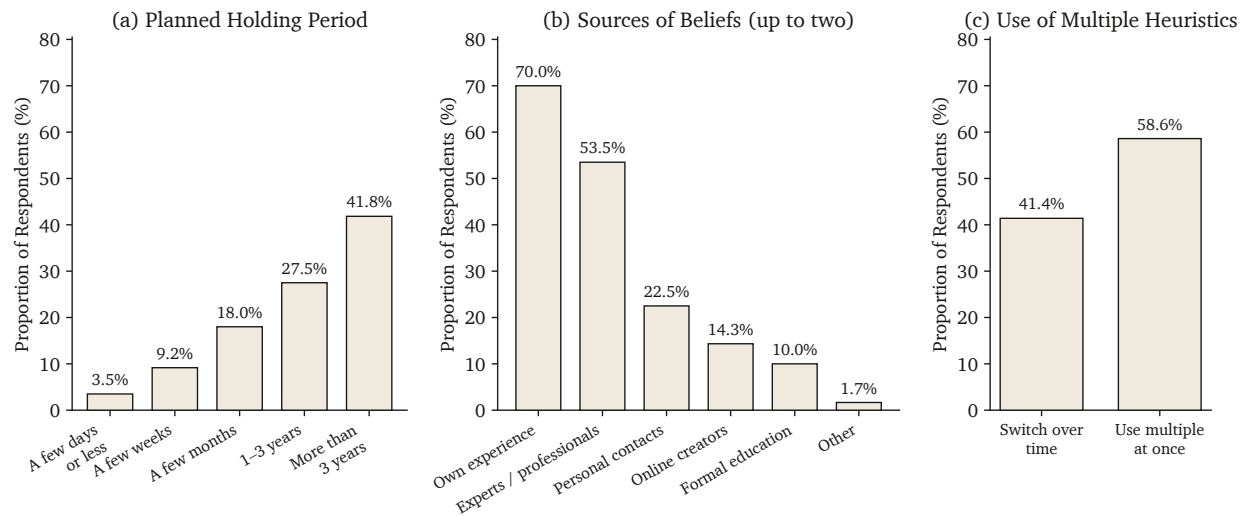


Figure D2. Planned Holding Periods, Sources of Beliefs About Heuristic Effectiveness, and Use of Multiple Heuristics

This figure shows the distribution of responses to Questions 3 through 5 of the follow-up survey (Online Appendix D.1). Each bar and the number above it show the share of respondents who select the answer choice. Panels (a) and (b) cover all 600 respondents. In Panel (b), respondents select up to two sources of their belief that their approaches work in their country's stock market, so the shares sum to more than 100%, and we order sources by their share. Panel (c) covers the 529 respondents with more than one heuristic identified in the interview. The other respondents do not receive this question.

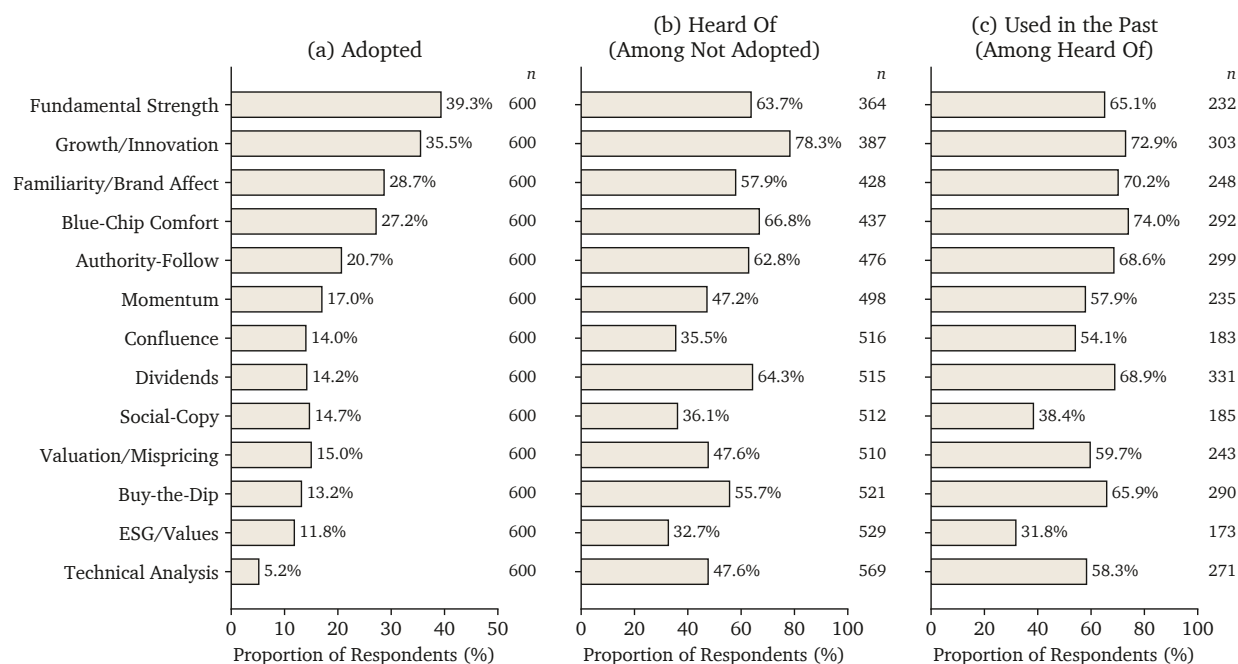


Figure D3. Adoption, Awareness, and Past Use of Heuristics

This figure shows the responses to Questions 6 and 7 of the follow-up survey (Online Appendix D.1), together with the heuristics each respondent adopts. Panel (a) shows, among the 600 follow-up survey respondents, the share who adopt each heuristic, that is, for whom it is among the heuristics we identify in their interview. Panel (b) covers the heuristics a respondent has not adopted and shows the share of respondents who have heard of the heuristic (6,262 respondent–heuristic pairs). Panel (c) shows, among respondents who have heard of a heuristic, the share who have used it in the past (3,285 pairs). In each panel, the column to the right of the bars reports the number of respondents behind each bar. Eleven of the 600 respondents select “None of the above” in Question 6, and 13 of the remaining 589 do so in Question 7. We order heuristics from top to bottom by their prevalence in the full interview sample.

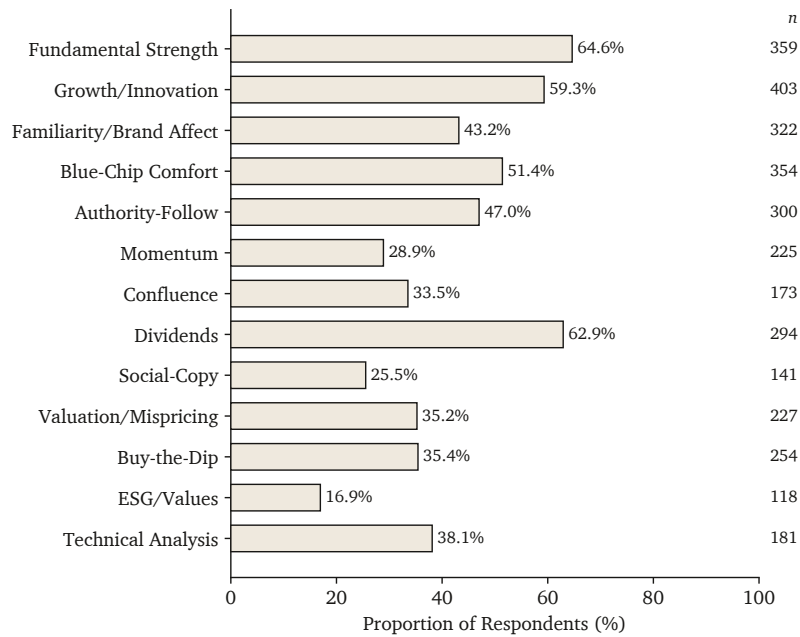


Figure D4. Three Longest-Used Heuristics out of Those Adopted or Previously Used

This figure shows the responses to Question 8 of the follow-up survey (Online Appendix D.1). Only the 512 respondents whose adopted and previously used heuristics number more than three receive the question. Each of them selects the three used the longest, and the figure shows the share of respondents selecting a heuristic when it is among their options (3,351 respondent–heuristic pairs). The column to the right of the bars reports the number of respondents behind each bar. We order heuristics from top to bottom by their prevalence in the full interview sample.

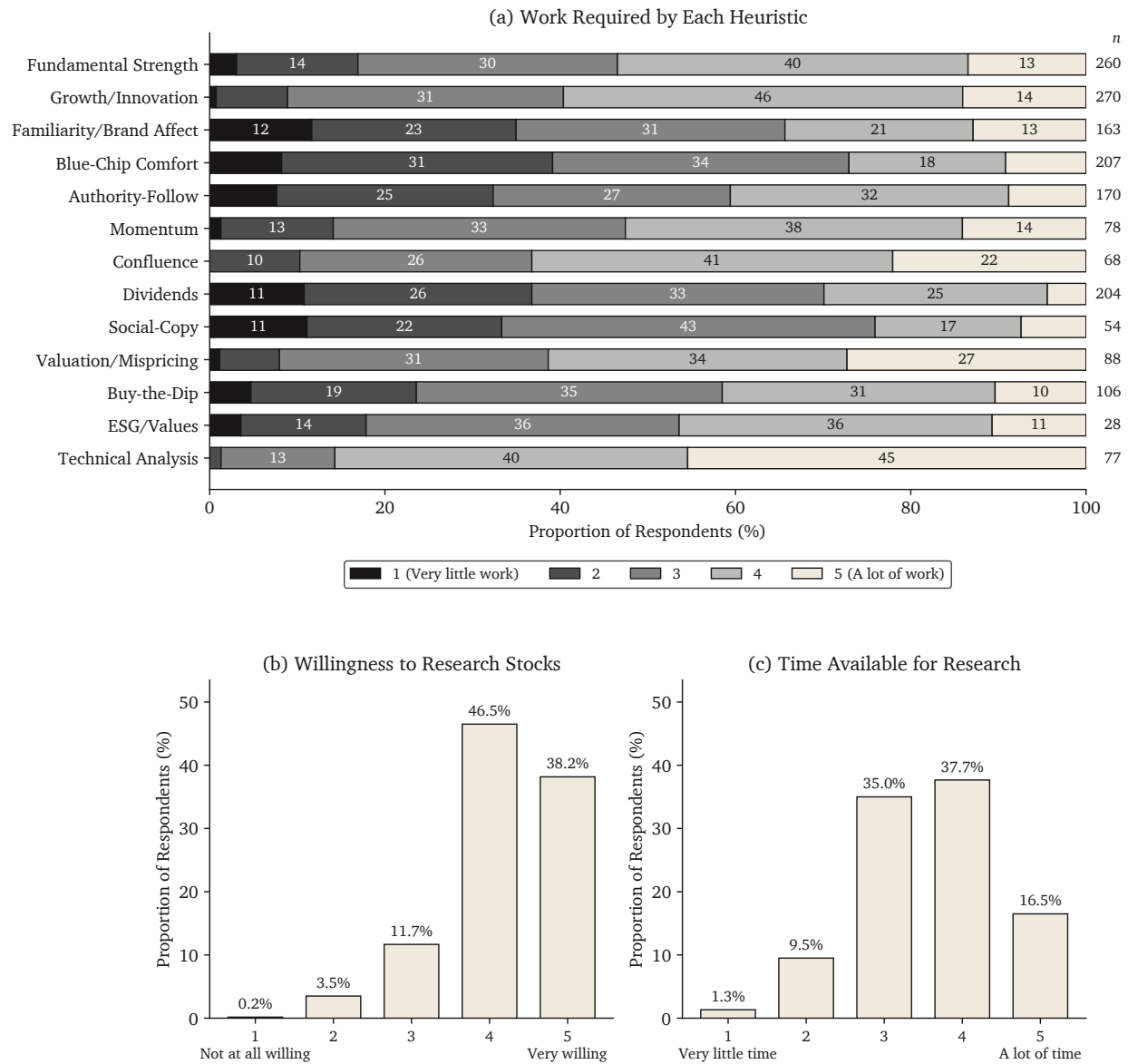


Figure D5. Perceived Work Required by Each Heuristic, Willingness to Research Stocks, and Time Available for Research

This figure shows the distribution of responses to Questions 9, 12, and 13 of the follow-up survey (Online Appendix D.1). Panel (a) shows how much work respondents believe each heuristic requires, on a five-point scale from 1 (very little work) to 5 (a lot of work). Respondents rate only the heuristics they have used the longest (at most three), which yields 1,773 respondent–heuristic pairs. The column to the right of the bars reports the number of respondents behind each bar. We print shares inside the bars when they are at least 10%. Because we round these shares to whole percentages, they may not sum to 100%. We order heuristics from top to bottom by their prevalence in the full interview sample. Panels (b) and (c) cover all 600 respondents and show their willingness to research stocks before buying and the time they have available for such research, each on a five-point scale.

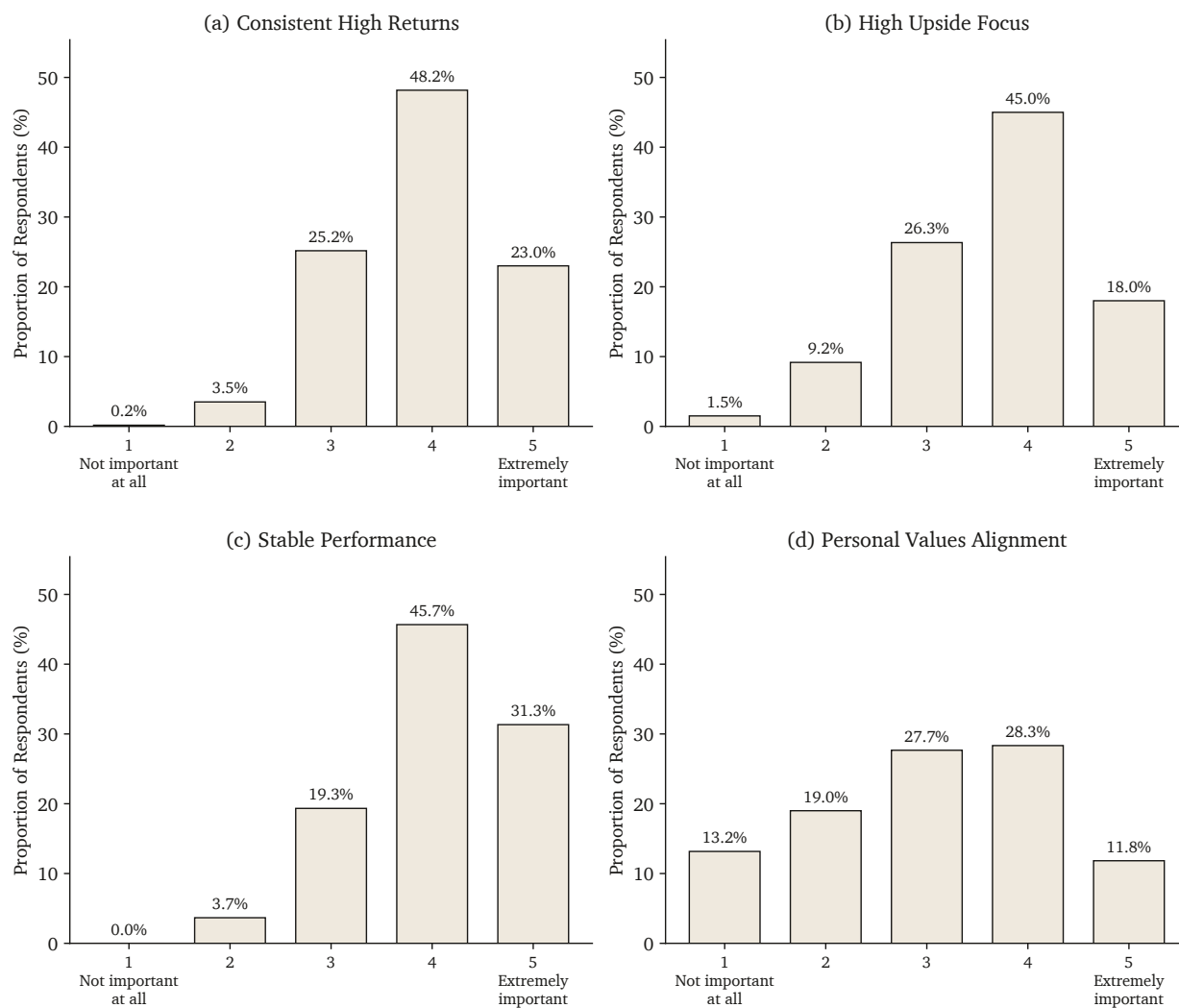


Figure D6. Personal Importance of the Four Investment Objectives

This figure shows the distribution of responses to Question 10 of the follow-up survey (Online Appendix D.1). All 600 respondents rate the importance of each of four investment objectives on a five-point scale from 1 (not important at all) to 5 (extremely important). Each panel shows one objective, and each bar and the number above it show the share of respondents who give the rating.

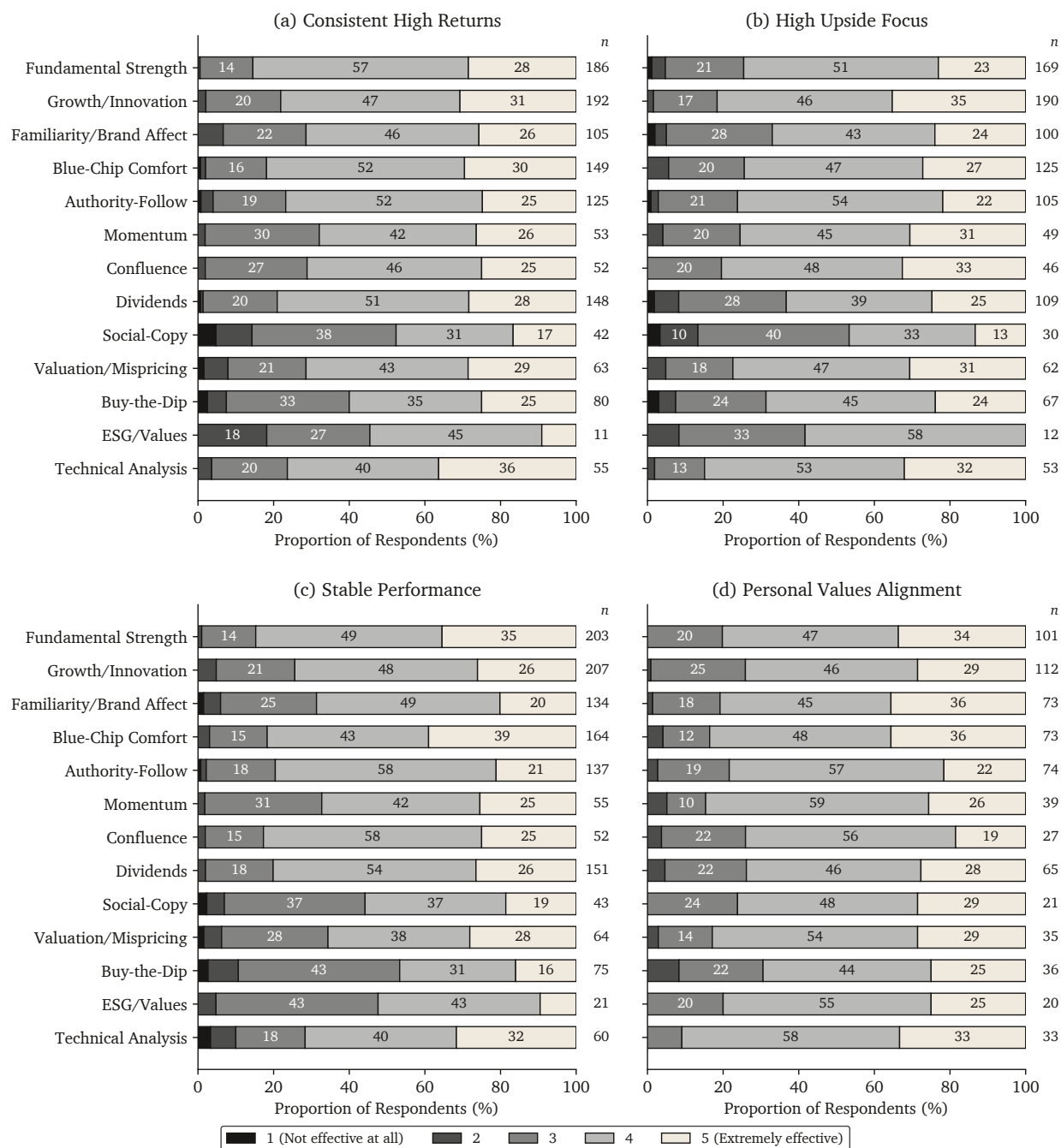


Figure D7. Distribution of Perceived Effectiveness Ratings, by Objective

This figure shows the distribution of responses to Question 11 of the follow-up survey (Online Appendix D.1). Respondents rate how good a heuristic is at finding stocks that match an objective in their country's stock market, on a five-point scale from 1 (not effective at all) to 5 (extremely effective). They rate only the heuristics they have used the longest (at most three) and only for the objectives they rate 4 or 5 in importance. The panels for consistent high returns, high upside focus, stable performance, and personal values alignment cover 1,261, 1,117, 1,366, and 709 respondent-heuristic pairs, respectively, and the column to the right of the bars reports the number of respondents behind each bar. We print shares inside the bars when they are at least 10%. Because we round these shares to whole percentages, they may not sum to 100%. We order heuristics from top to bottom by their prevalence in the full interview sample.

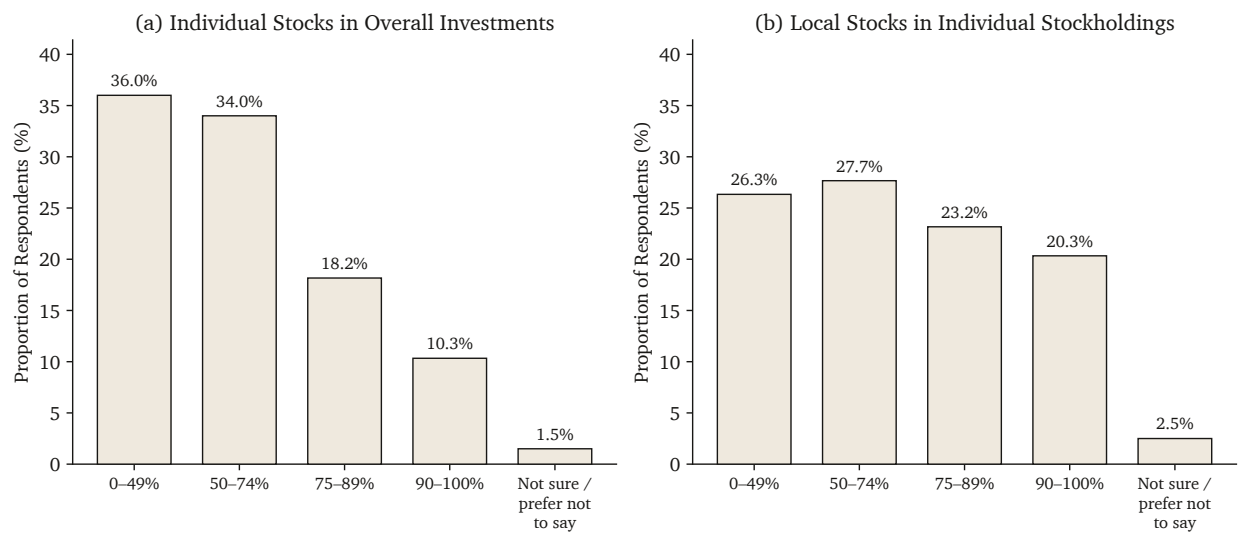


Figure D8. Portfolio Shares in Individual Stocks and in Local Stocks

This figure shows the distribution of responses to Questions 14 and 15 of the follow-up survey (Online Appendix D.1). Panel (a) shows the share of respondents' overall investments held in individual stocks, and Panel (b) shows the share of their individual stockholdings invested in locally listed stocks. Each bar and the number above it show the share of the 600 respondents who select the answer choice.

Online Appendix E. Robustness to Alternative Mappings between Heuristics and Firm Characteristics

Our measures of heuristic effectiveness depend on the firm characteristics we assign to each heuristic in Table 5 of the main paper. Because several firm characteristics can be related to the same heuristic, these assignments involve judgment. We therefore construct alternative mappings with LLMs, recalculate heuristic effectiveness under each mapping, and repeat our main tests.

We let LLMs stand in for the investors who use each heuristic. For one heuristic at a time, an LLM chooses the firm characteristics it would use to select stocks, either as a generic retail investor or with the background of an interviewed investor who uses that heuristic. It never sees our mapping or our results. We collect 10,112 such responses from two LLMs and two prompts and combine them into two kinds of alternative mappings, a consensus mapping for each LLM and prompt that keeps the most common choices and 7,596 distinct mappings that each combine one response per heuristic. We then ask how often our main conclusions survive when we replace our mapping with each alternative. The four steps below describe the procedure in detail.

1. **Inputs and models.** We use GPT-6.1 Sol and Claude Sonnet 5.5, both with high reasoning effort.⁷ Each call supplies the description of one heuristic from Table 3 of the main paper and a list of candidate characteristics. For each candidate, the model sees only the variable name, such as *ni_be*, and a definition based on JKP’s documentation of factors. JKP group their 153 factors into 13 clusters, and we draw the candidates for each heuristic from the clusters related to it. We exclude composite scores and heavily model-based measures because retail investors are unlikely to use them. For *Buy-the-Dip* and *Dividends*, whose clusters offer too few relevant candidates, we add a few closely related characteristics from the “Other Factors” list in the JKP documentation. The candidate sets are as follows.

- For *Fundamental Strength*, we use the Profitability, Quality, Low Leverage, and Size clusters without the composite scores *f_score*, *o_score*, *z_score*, *qmj*, *qmj_prof*, *qmj_growth*, and *qmj_safety* (37 candidates).
- For *Growth/Innovation*, we use the Investment, Profit Growth, and Size clusters (39

⁷The model identifiers are gpt-6.1-sol and claude-sonnet-5-5. Both were among the most recent models available when we conducted this analysis.

candidates).

- For *Blue-Chip Comfort*, we use the Low Leverage, Low Risk, and Size clusters without the composite score z_score (33 candidates).
- For *Momentum*, we use the Momentum cluster (8 candidates).
- For *Buy-the-Dip*, we use the Short-Term Reversal cluster and add the past two- and three-month returns, ret_2_0 and ret_3_0 (8 candidates).
- For *Dividends*, we use the four payout characteristics of the Value cluster, $div12m_me$, $eqnpo_12m$, $eqnpo_me$, and $eqpo_me$, and add the trailing one-, three-, and six-month dividend yields, $div1m_me$, $div3m_me$, and $div6m_me$ (7 candidates).
- For *Valuation/Mispricing*, we use the Value cluster without the model-based measures eq_dur and $ival_me$ (16 candidates).

The candidate sets span 124 distinct characteristics in the JKP global factor data and include every characteristic in our mapping.

2. **Prompts and LLM calls.** We use two prompts. Prompt A asks the model to act as a retail investor who applies the heuristic and not to invent personal characteristics. Prompt B adds the background of an investor we interviewed who adopts the heuristic, and tells the model to use this background as context for what the investor could understand and apply. Conditioning a language model on the background of real respondents brings its answers closer to those of the corresponding human subgroups (Argyle, Busby, Fulda, Gubler, Rytting, and Wingate 2023) and diversifies them, so we can test our results on a wider range of maps. The background covers country, gender, age group, education, investment knowledge, investable assets, and risk attitude. For Prompt B, we make one call for each pair of an investor and a heuristic the investor adopts, so that the responses mirror the actual users of each heuristic. There are 2,528 such pairs, because many investors adopt more than one.⁸ We make the same number of Prompt A calls for each heuristic. Each LLM therefore answers 2,528 calls per prompt, for 10,112 responses in total. Every call starts a new independent session. The instructions tell the model to base its choice only on the supplied information, without outside information or

⁸The numbers of adopters are 623 for *Fundamental Strength*, 579 for *Growth/Innovation*, 433 for *Blue-Chip Comfort*, 276 for *Momentum*, 192 for *Buy-the-Dip*, 216 for *Dividends*, and 209 for *Valuation/Mispricing*. These counts correspond to the prevalence of each heuristic in Table 3 of the main paper.

prior knowledge.

3. **Responses and maps.** Each response selects one to three characteristics, each of which can be used to screen or rank stocks for the heuristic. Combining one response from each of the seven heuristics in every possible way, we get 144, 360, 2,340, and 5,616 distinct maps for GPT Prompt A, GPT Prompt B, Claude Prompt A, and Claude Prompt B, respectively, and 7,596 distinct maps across the four setups. We count each distinct map once, however often the LLMs propose it, because our goal is to examine how much our findings depend on which plausible mapping a researcher might adopt. To summarize the typical choices of each LLM and prompt, we also form a consensus map that keeps, for each heuristic, the characteristics the responses select most often. We provide the four consensus mappings in Table E1.
4. **Repeating tests.** For each distinct map, we recalculate heuristic effectiveness and rerun the main tests exactly as in the main paper.

In Figure E1, we compare the estimates under every possible map with those under our mapping. Every map yields positive belief-accuracy slopes and positive historical delivery, whereas the objective-specific out-of-sample estimates spread more widely and fall below zero for a few maps. In Figure E2, we show the corresponding p -values. Apart from the returns slope of belief accuracy, which is not significant under our mapping either, p -values exceed 0.10 only for the out-of-sample tests for returns and stability, and only under some maps.

In Table E2, we report the percentage of distinct maps in each setup that support each main-paper result, which shows how sensitive our results are to the choice of map. A map supports a result if its estimate matches the corresponding main-paper estimate in sign and in statistical significance at the 10% level. A map also supports a result if its estimate favors our hypothesis more strongly by being significantly positive where the main-paper estimate is positive but not significant.

For belief accuracy and historical delivery, every map supports the pooled result and the result for each objective. In 640 of the 7,596 maps, the returns slope of belief accuracy even becomes significant. Out-of-sample delivery depends more on the mapping. Every map supports the pooled and upside results, whereas, depending on the setup, 83% to 100% of the maps support the returns result and 58% to 89% support the stability result. In the remaining maps, the estimate is positive but not significant, except in 79 Claude-based maps, where it is negative but not significant. Thus,

belief accuracy, historical delivery, and pooled out-of-sample delivery do not depend on our choice of characteristics, whereas out-of-sample delivery for returns and especially stability is more sensitive to it.

Table E1. Mapping Heuristics to Firm Characteristics: LLM Consensus

This table reports the consensus mapping of each model–prompt setup. For each heuristic, the consensus keeps the characteristics that the responses select most often, and the number it keeps is the most common number of characteristics per response. Panel A uses GPT-6.1 Sol and Panel B uses Claude Sonnet 5.5. Prompt A supplies no investor background, and Prompt B adds the background of an interviewed investor who adopts the heuristic. Bold variable names mark characteristics that our mapping assigns to the same heuristic, and a blank cell means that the consensus for that prompt does not include the characteristic. As in Table 5 of the main paper, a positive (negative) sign means that we take the heuristic to favor stocks with high (low) values of the characteristic. Within each heuristic, the bold characteristics come first, in the order of that table, and the others follow in alphabetical order of their variable names. We give the models definitions based on the JKP documentation of factors, so their wording can differ from that in this table, which is polished for readability.

Panel A: GPT-Based Mapping

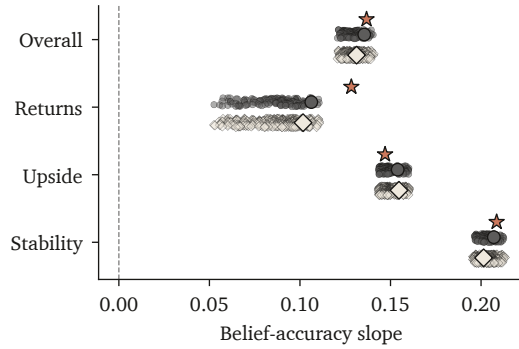
Heuristic	Characteristic (sign)		Definition
	Prompt A	Prompt B	
Fundamental Strength	<i>ni_be</i> (+)	<i>ni_be</i> (+)	Return on equity (net income / book equity)
	<i>ebit_sale</i> (+)	<i>ebit_sale</i> (+)	Operating margin (EBIT / sales)
	<i>ocf_at</i> (+)	<i>ocf_at</i> (+)	Operating cash-flow profitability (operating cash flow / assets)
Growth/Innovation	<i>rd_me</i> (+)	<i>rd_me</i> (+)	R&D-to-market ratio (R&D expense / market equity)
	<i>sale_gr3</i> (+)	<i>sale_gr3</i> (+)	Sales growth (3 years)
Blue-Chip Comfort	<i>me</i> (+)	<i>me</i> (+)	Market equity
	<i>age</i> (+)	<i>age</i> (+)	Firm age
	<i>rvol_21d</i> (–)	<i>rvol_21d</i> (–)	Return volatility (21 days)
Buy-the-Dip	<i>ret_1_0</i> (–)	<i>ret_1_0</i> (–)	Past 1-month return
	<i>ret_2_0</i> (–)	<i>ret_2_0</i> (–)	Past 2-month return
	<i>ret_3_0</i> (–)	<i>ret_3_0</i> (–)	Past 3-month return
Dividends	<i>div12m_me</i> (+)	<i>div12m_me</i> (+)	Dividend yield (trailing 12-month)
Momentum	<i>ret_6_1</i> (+)	<i>ret_6_1</i> (+)	6-month momentum (return $t-6 \rightarrow t-1$)
	<i>ret_9_1</i> (+)	<i>ret_9_1</i> (+)	9-month momentum (return $t-9 \rightarrow t-1$)
	<i>ret_12_1</i> (+)	<i>ret_12_1</i> (+)	12-month momentum (return $t-12 \rightarrow t-1$)
Valuation/Mispricing	<i>be_me</i> (+)	<i>be_me</i> (+)	Book-to-market ratio (book equity / market equity)
	<i>bev_mev</i> (+)	<i>bev_mev</i> (+)	Book-to-market enterprise value (book enterprise value / market enterprise value)
		<i>fcf_me</i> (+)	Free-cash-flow yield (free cash flow / market equity)

Table E1. Mapping Heuristics to Firm Characteristics: LLM Consensus (continued)

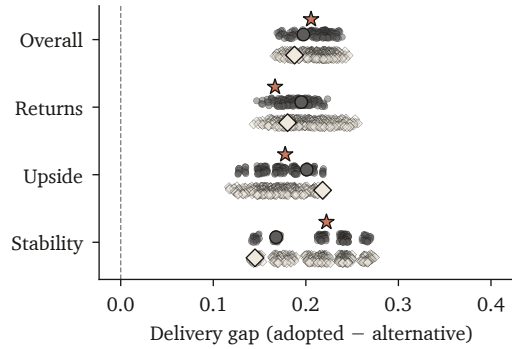
Panel B: Claude-Based Mapping

Heuristic	Characteristic (sign)		Definition
	Prompt A	Prompt B	
Fundamental Strength		ni_be (+)	Return on equity (net income / book equity)
	<i>cop_at</i> (+)	<i>cop_at</i> (+)	Cash-based operating profitability ((EBITDA + R&D expense – operating accruals) / assets)
	<i>niq_be</i> (+)		Quarterly return on equity (quarterly net income / lagged book equity)
	<i>ocf_at</i> (+)	<i>ocf_at</i> (+)	Operating cash-flow profitability (operating cash flow / assets)
Growth/Innovation	rd_me (+)	rd_me (+)	R&D-to-market ratio (R&D expense / market equity)
	<i>capx_gr3</i> (+)	<i>capx_gr3</i> (+)	Capital-expenditure growth (3 years)
	<i>sale_gr3</i> (+)	<i>sale_gr3</i> (+)	Sales growth (3 years)
Blue-Chip Comfort	me (+)	me (+)	Market equity
	age (+)	age (+)	Firm age
		rvol_21d (–)	Return volatility (21 days)
	<i>ivol_capm_252d</i> (–)		CAPM idiosyncratic volatility (252 trading days)
Buy-the-Dip	ret_1_0 (–)	ret_1_0 (–)	Past 1-month return
	ret_2_0 (–)	ret_2_0 (–)	Past 2-month return
	ret_3_0 (–)	ret_3_0 (–)	Past 3-month return
Dividends	div6m_me (+)	div6m_me (+)	Dividend yield (trailing 6-month)
	div12m_me (+)	div12m_me (+)	Dividend yield (trailing 12-month)
		<i>div3m_me</i> (+)	Dividend yield (trailing 3-month)
Momentum	ret_6_1 (+)	ret_6_1 (+)	6-month momentum (return $t-6 \rightarrow t-1$)
	ret_12_1 (+)	ret_12_1 (+)	12-month momentum (return $t-12 \rightarrow t-1$)
	<i>prc_highprc_252d</i> (+)	<i>prc_highprc_252d</i> (+)	Price-to-52-week-high ratio (price / highest price over 252 trading days)
Valuation/Mispricing	be_me (+)	be_me (+)	Book-to-market ratio (book equity / market equity)
	<i>ebitda_mev</i> (+)	<i>ebitda_mev</i> (+)	EBITDA-to-enterprise-value ratio (EBITDA / market enterprise value)
	<i>fcf_me</i> (+)	<i>fcf_me</i> (+)	Free-cash-flow yield (free cash flow / market equity)

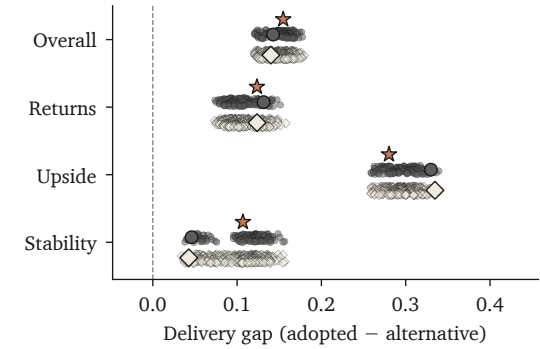
GPT-Based Mapping



(a) Belief Accuracy (Figure 8)

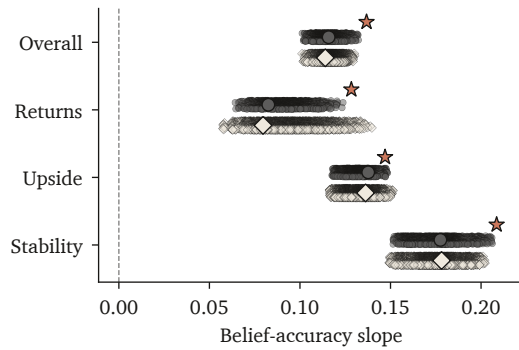


(b) Historical Delivery (Table 7)

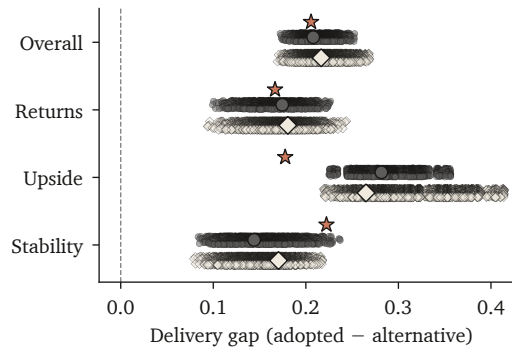


(c) Out-of-Sample Delivery (Table 8)

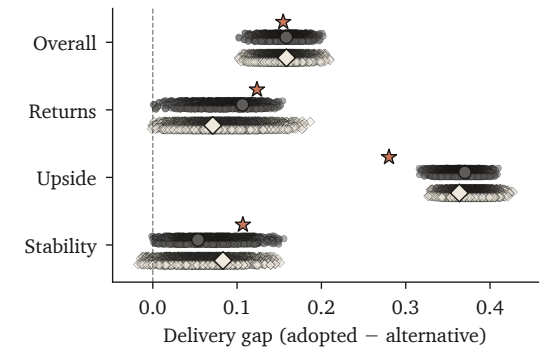
Claude-Based Mapping



(d) Belief Accuracy (Figure 8)



(e) Historical Delivery (Table 7)



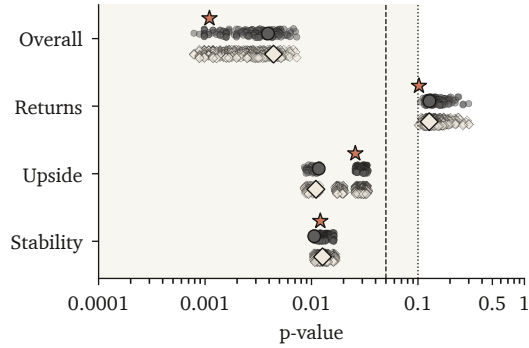
(f) Out-of-Sample Delivery (Table 8)

★ Our mapping ● Prompt A ◇ Prompt B

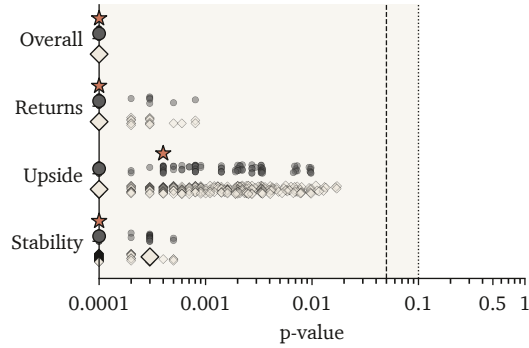
Figure E1. Sensitivity of Belief Accuracy and Heuristic Delivery to Alternative Mappings

This figure compares the estimates under every possible map with those under our mapping. Panels (a) through (c) use GPT-6.1 Sol and Panels (d) through (f) use Claude Sonnet 5.5. Panels (a) and (d) show the belief-accuracy slope of Figure 8 of the main paper. The other panels show the delivery gap, which is the *Adopted heuristics* row minus the *Alternative heuristics* row of Table 7 (historical) and Table 8 (out-of-sample). As in the main paper, we standardize effectiveness within each objective, here separately for each map. Each small symbol is one possible distinct map, the large symbols are the consensus maps, and stars mark our mapping. Prompt A (dark circles) supplies no investor background, and Prompt B (light diamonds) adds the background of an investor who adopts the heuristic. Vertical offsets within each row only separate the symbols, and the dashed line marks zero.

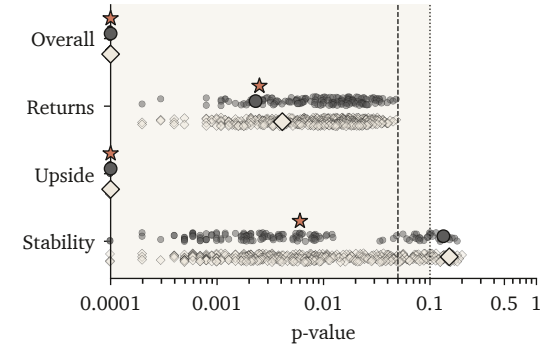
GPT-Based Mapping



(a) Belief Accuracy (Figure 8)

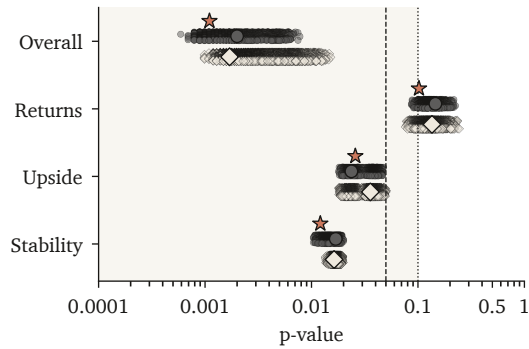


(b) Historical Delivery (Table 7)

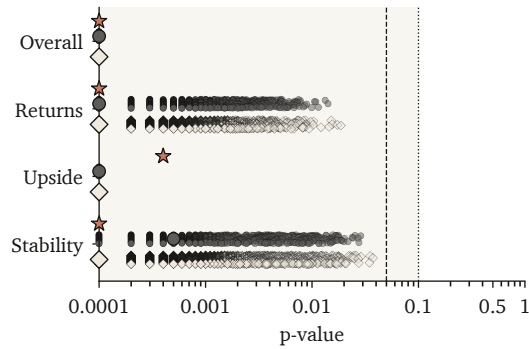


(c) Out-of-Sample Delivery (Table 8)

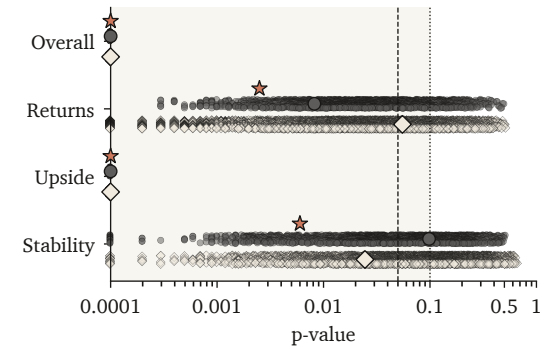
Claude-Based Mapping



(d) Belief Accuracy (Figure 8)



(e) Historical Delivery (Table 7)



(f) Out-of-Sample Delivery (Table 8)

★ Our mapping ● Prompt A ◇ Prompt B

Figure E2. Statistical Significance of Belief Accuracy and Heuristic Delivery across Alternative Mappings

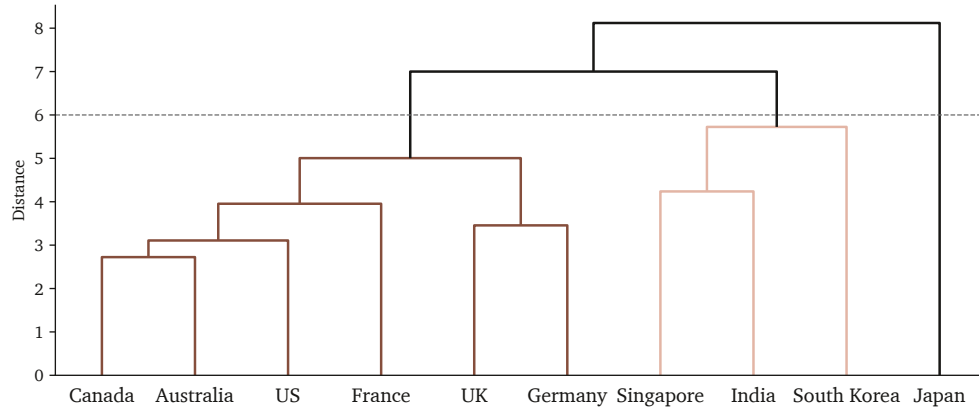
This figure shows the p -values of the estimates in Figure E1. Panels (a) through (c) use GPT-6.1 Sol and Panels (d) through (f) use Claude Sonnet 5.5. As in the main paper, belief accuracy uses the two-sided wild-cluster bootstrap p -value by country, and delivery uses the one-sided p -value against equally sized random draws from the alternative heuristics. The horizontal axis is logarithmic, the shaded area marks p -values below 0.10, and the dashed and dotted lines mark 0.05 and 0.10. We show p -values below 0.0001, including zero, at 0.0001. Each small symbol is one possible distinct map, the large symbols are the consensus maps, and stars mark our mapping. Prompt A (dark circles) supplies no investor background, and Prompt B (light diamonds) adds the background of an investor who adopts the heuristic. Vertical offsets within each row only separate the symbols.

Table E2. Support for Main-Paper Results across Alternative Mappings

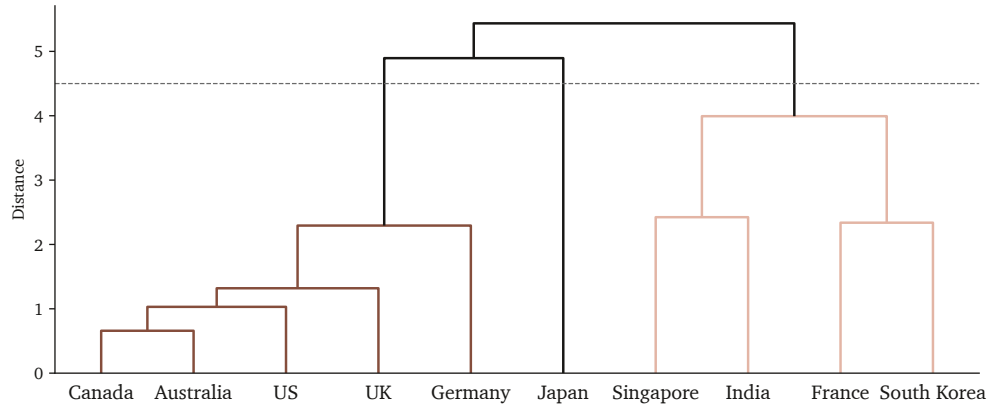
This table reports, for each model–prompt setup, the percentage of the N possible distinct maps that support each main-paper result. We count each distinct map once. A map supports a result if its estimate matches the corresponding main-paper estimate in sign and in whether it is significant at the 10% level, or favors our hypothesis more strongly. GPT-based mapping uses GPT-6.1 Sol and Claude-based mapping uses Claude Sonnet 5.5. Prompt A supplies no investor background, and Prompt B adds the background of an investor who adopts the heuristic. As in the main paper, belief accuracy uses the two-sided wild-cluster bootstrap p -value by country, and delivery uses the one-sided p -value against equally sized random draws from the alternative heuristics.

Main-paper result	Analysis	Objective	GPT-based mapping		Claude-based mapping	
			Prompt A $N = 144$	Prompt B $N = 360$	Prompt A $N = 2,340$	Prompt B $N = 5,616$
Figure 8	Belief accuracy	Overall	100%	100%	100%	100%
		Returns	100%	100%	100%	100%
		Upside	100%	100%	100%	100%
		Stability	100%	100%	100%	100%
Table 7	Historical delivery	Overall	100%	100%	100%	100%
		Returns	100%	100%	100%	100%
		Upside	100%	100%	100%	100%
		Stability	100%	100%	100%	100%
Table 8	Out-of-sample delivery	Overall	100%	100%	100%	100%
		Returns	100%	100%	87%	83%
		Upside	100%	100%	100%	100%
		Stability	89%	89%	71%	58%

Online Appendix F. Additional Figures and Tables



(a) By Heuristic Prevalence



(b) By Culture Scores

Figure F1. Country Similarity in Heuristic Prevalence and Culture

This figure shows hierarchical clustering dendrograms that group the ten countries in our sample under two measures of similarity. In Panel (a), we represent each country by the proportions of its investors who invoke each of the thirteen heuristics. An investor invokes a heuristic if its score is at least three (“meaningful contributor”). In Panel (b), we represent each country by its six Hofstede culture scores (power distance, individualism, masculinity, uncertainty avoidance, long-term orientation, and indulgence). The six-dimension framework is from Hofstede, Hofstede, and Minkov (2010), and we obtain the country scores from The Culture Factor Group (<https://www.theculturefactor.com/country-comparison-tool>), whose individualism and long-term-orientation scores are the revisions of Minkov and Kaasa (2022) rather than the values published in Hofstede, Hofstede, and Minkov (2010). In both panels, we standardize each element of the vector across the ten countries and cluster the countries agglomeratively using complete linkage with Euclidean distance. The vertical axis reports the height at which two clusters merge, that is, the largest Euclidean distance between any pair of countries in the two merged clusters. Greater height indicates greater dissimilarity. The dashed horizontal line marks the threshold that defines the clusters, 6.0 in Panel (a) and 4.5 in Panel (b). We draw branches that merge below the threshold in cluster-specific shades and branches above it in black.

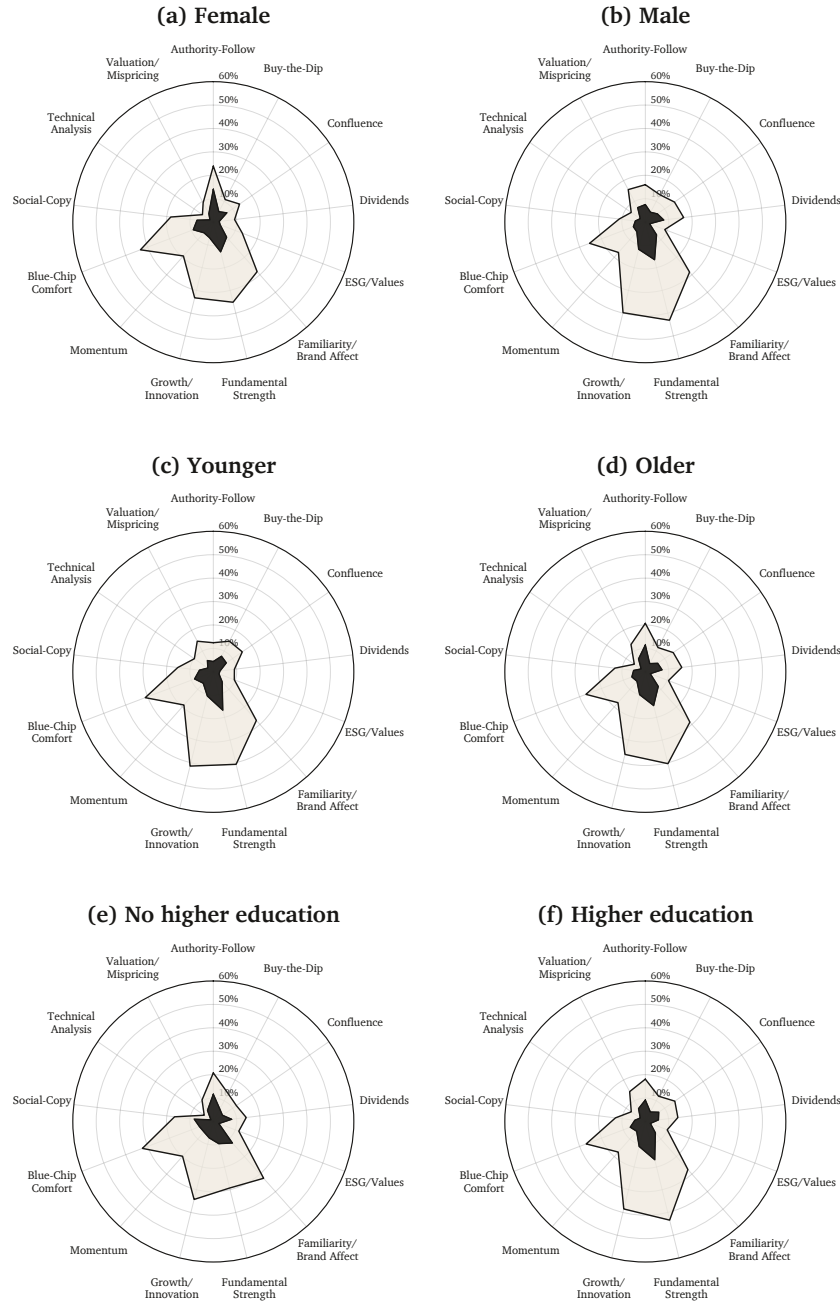


Figure F2. Heuristic Prevalence by Investor Demographics

This figure shows, separately by investor demographic group, the proportion of investors who invoke each of the thirteen identified recurrent heuristics. We split the investors into two groups along each demographic dimension. This page reports the splits by gender (*Female* and *Male*), age (*Older*, age ≥ 35 , and *Younger*), and education (*Higher education*, bachelor's degree or above, and *No higher education*). The gender split omits the single investor who reports another gender. The next page reports the splits by wealth, knowledge, and risk attitude. Within each panel we arrange the thirteen heuristics radially, with the distance from the center representing the proportion of investors in that group who invoke the heuristic. The dark region marks the proportion for whom the heuristic scores four ("primary driver"), the light region the proportion for whom it scores at least three ("meaningful contributor"). The dark region is therefore nested within the light one. The radial axis runs from 0% at the center to 60% at the outer circle, with concentric circles at 10-percentage-point intervals.

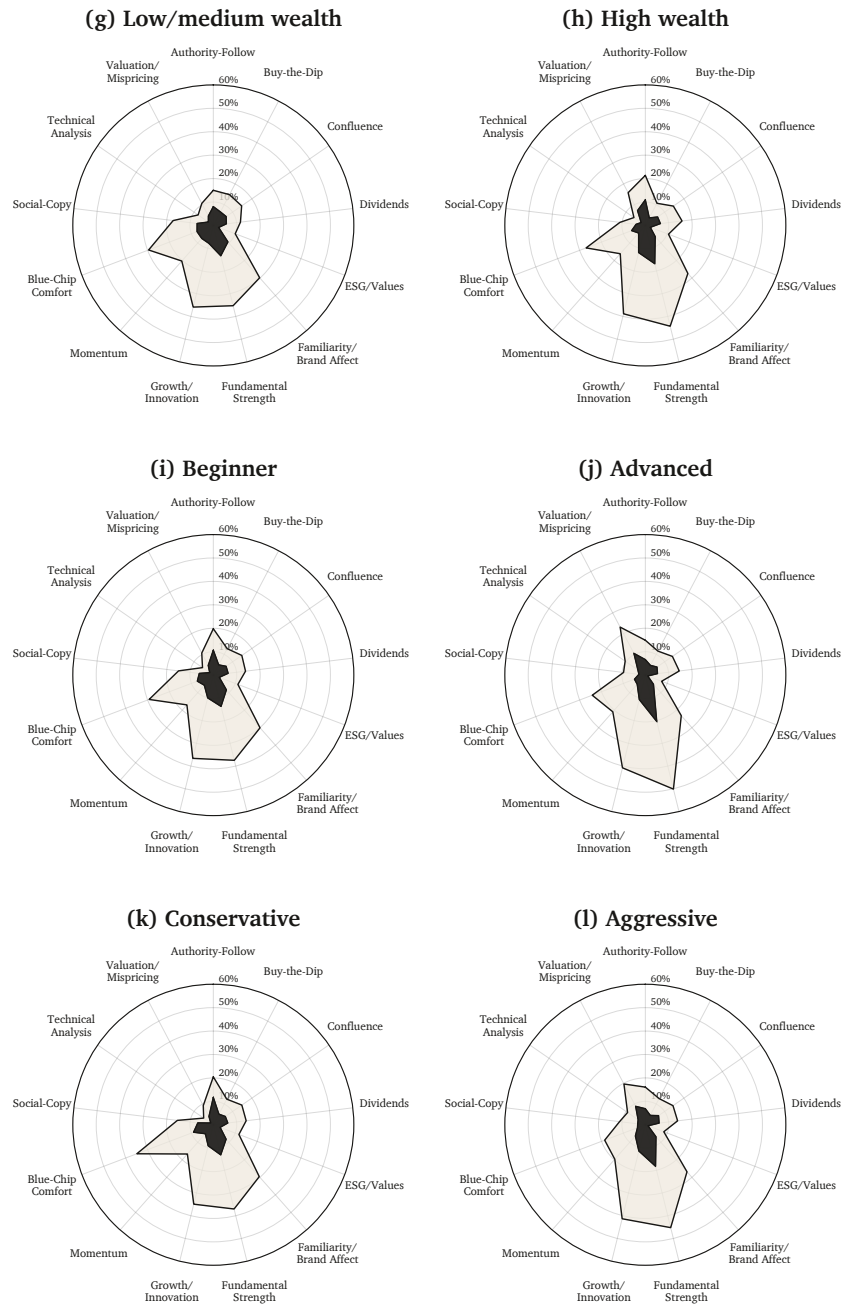
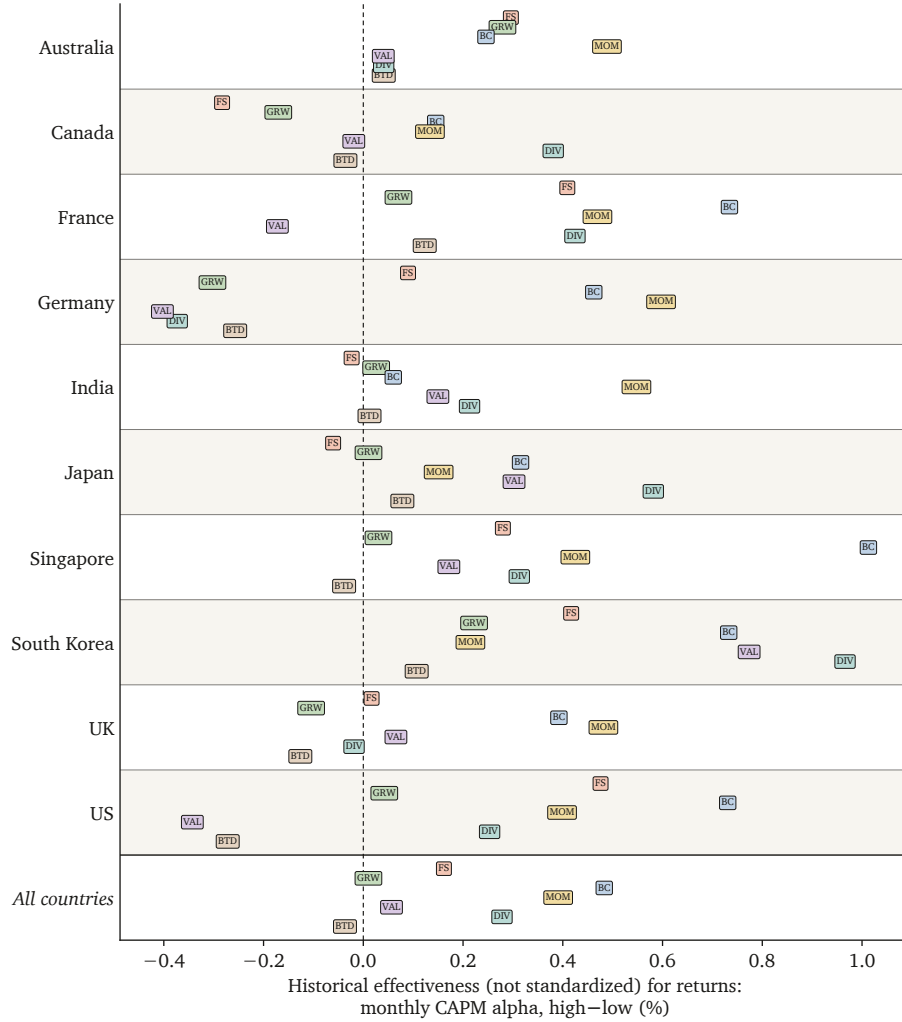


Figure F2. Heuristic Prevalence by Investor Demographics (continued)

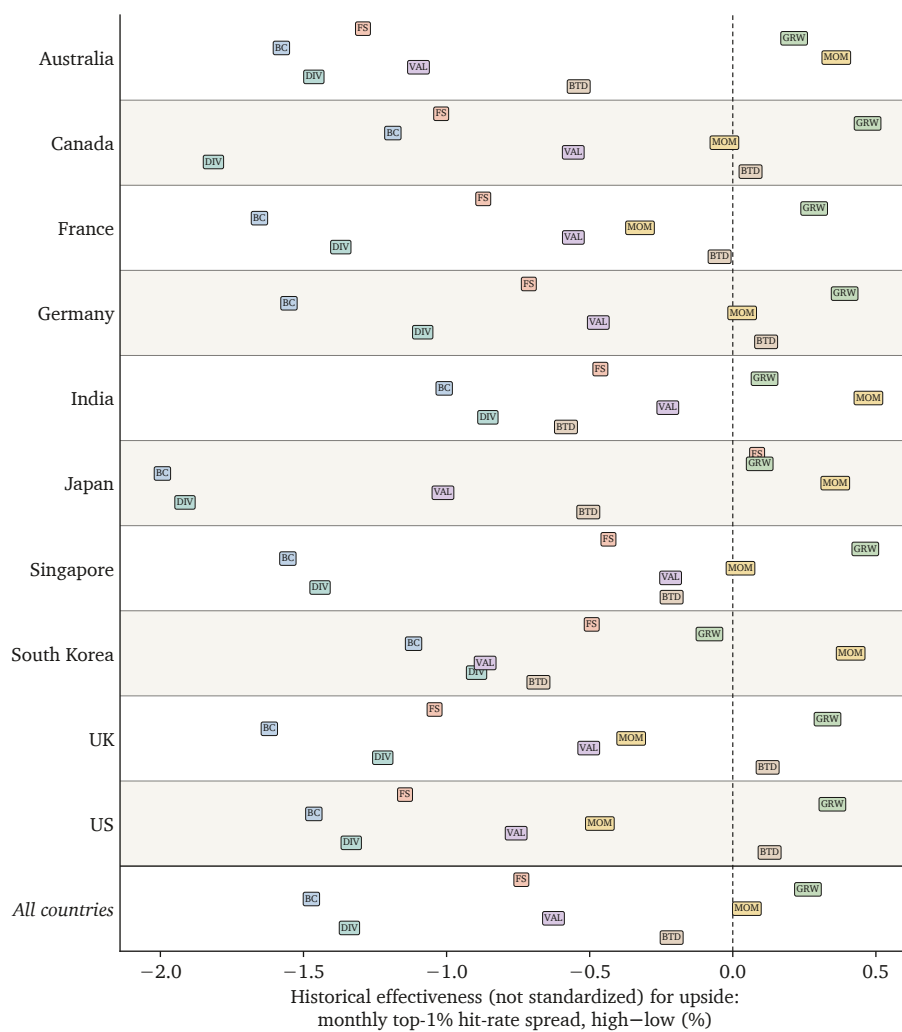
This figure continues Figure F2, reporting the splits by wealth (*High wealth*, investable assets $\geq$ USD 500,000, and *Low/medium wealth*), knowledge (*Advanced*, self-reported advanced or expert investment knowledge, and *Beginner*, complete beginner through intermediate), and risk attitude (*Aggressive*, self-reported willingness to take above-average or substantial financial risks, and *Conservative*, willingness to take no or average financial risks).



(a) Returns

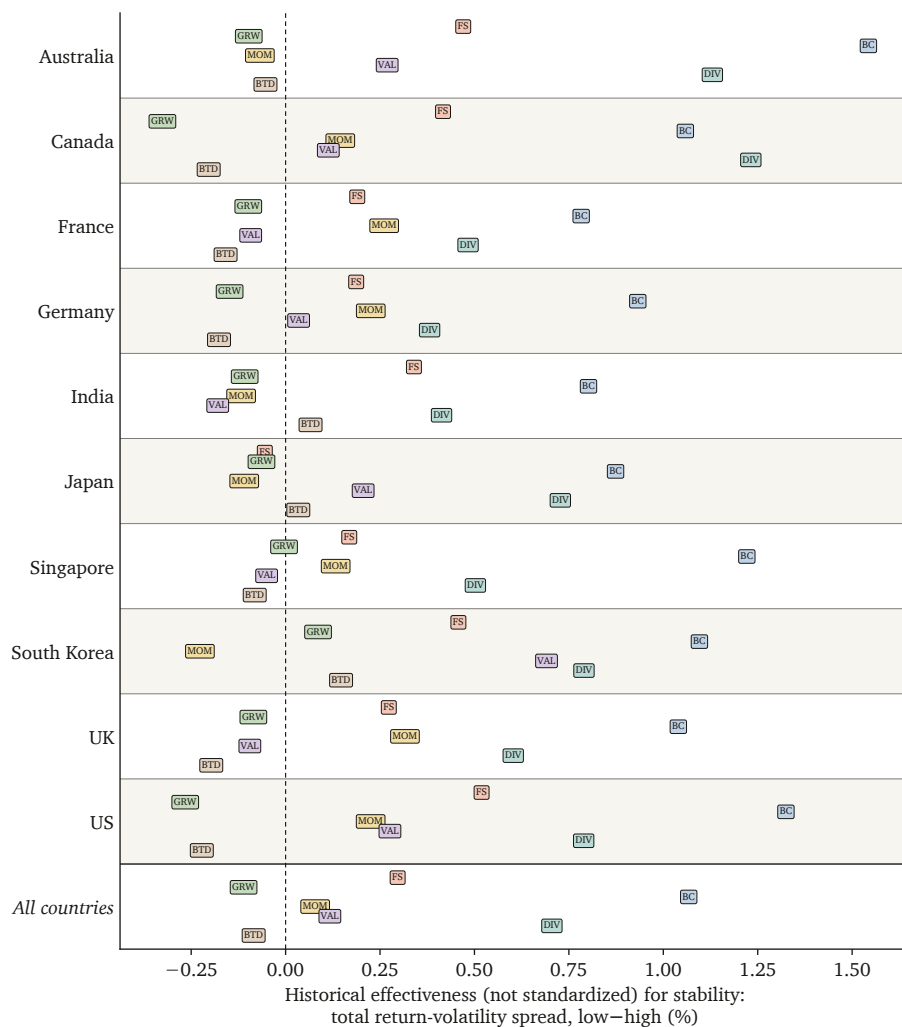
Figure F3. Historical Heuristic Delivery by Country and Objective

This figure shows the delivery of each of the seven heuristics covered by archival data on each of the three financial objectives, separately for each of the ten countries. We report delivery in native units, that is, before the within-objective standardization that turns delivery into historical effectiveness η_{hgc}^H (Section 5.3 of the main paper). As in Table 5 of the main paper, the delivery of a heuristic averages the outcome across the long-short portfolios formed on its three characteristics. Panel (a) reports returns delivery, the CAPM alpha of the value-weighted long-short portfolio (percent per month). Panel (b) reports upside delivery, the long-minus-short difference in the equally weighted fraction of stocks whose returns over the following month fall in the top 1% of the country-month cross-section (percentage points). Panel (c) reports stability delivery, the long-minus-short difference in the value-weighted average total return volatility over the following month (percent), with the sign reversed so that higher values indicate greater stability. All three panels use the same characteristic sorts over July 2015 through June 2025, the ten years preceding our interviews, and further right means better delivery. We draw each observation as the two- or three-letter code of its heuristic, namely FS (*Fundamental Strength*), GRW (*Growth/Innovation*), BC (*Blue-Chip Comfort*), MOM (*Momentum*), VAL (*Valuation/Mispricing*), DIV (*Dividends*), and BTD (*Buy-the-Dip*). The dashed vertical line marks zero delivery. Chip color and vertical position within each country band vary by heuristic for legibility and carry no information beyond the code. The *All countries* row reports the simple mean for each heuristic and objective across the ten countries.



(b) Upside

Figure F3. Historical Heuristic Delivery by Country and Objective (continued)
 This figure continues Figure F3, reporting upside delivery. See the notes to that figure for definitions.



(c) Stability

Figure F3. Historical Heuristic Delivery by Country and Objective (continued)

This figure continues Figure F3, reporting stability delivery. See the notes to that figure for definitions.

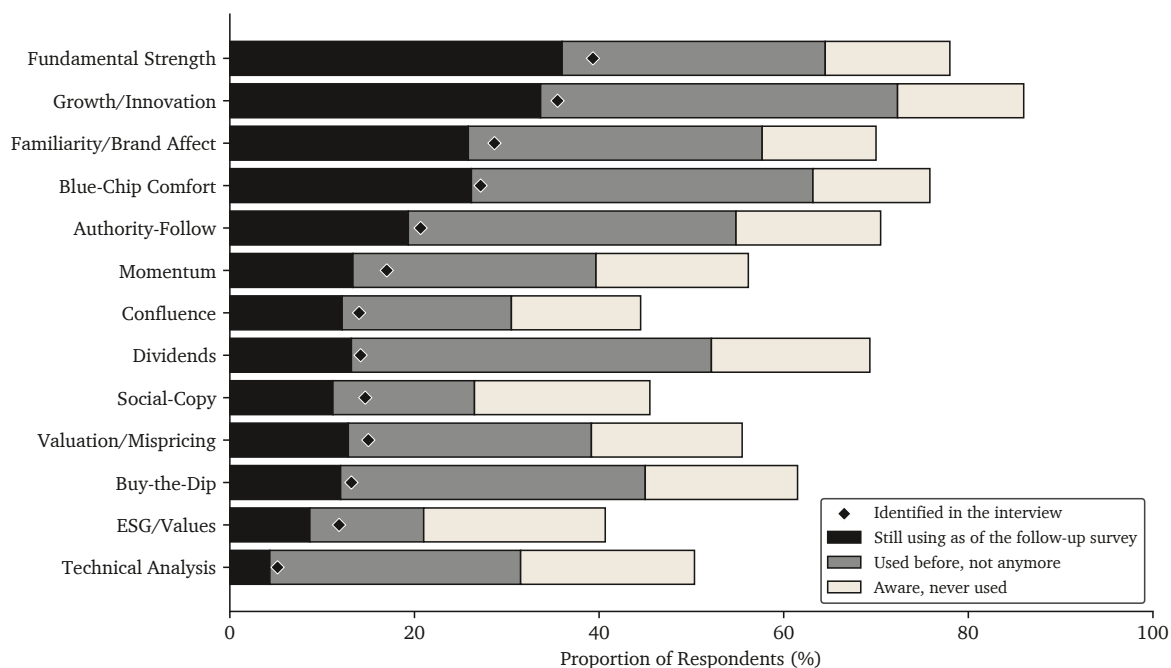


Figure F4. Heuristic Awareness and Use

This figure shows, for each of the thirteen heuristics, the proportions of follow-up survey respondents who use it, used it in the past, or have heard of it. The diamond marks the proportion of respondents in whose interview we identify the heuristic. The dark shading marks respondents in whose interview we identify the heuristic and who still use it at the time of the follow-up survey. The medium shading marks respondents who used the heuristic in the past but no longer do, and the light shading respondents who have heard of it but have never used it. The right end of each bar therefore marks the proportion of respondents aware of the heuristic. We order heuristics from top to bottom by their prevalence in the full interview sample.

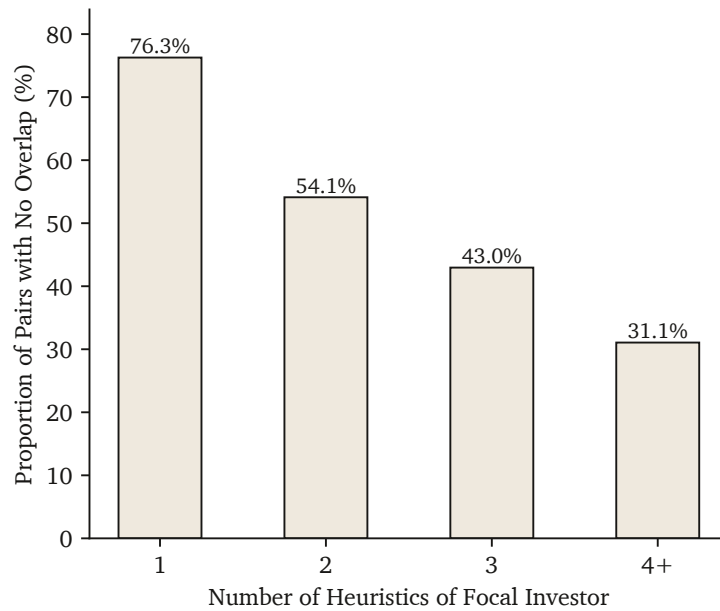


Figure F5. Proportion of Investor Pairs with Non-Overlapping Heuristics

This figure shows the proportion of investor pairs whose two members share no heuristic. An investor invokes a heuristic if its score is at least three (“meaningful contributor”). We form all 1,185,030 unordered pairs of the 1,540 investors we interviewed and classify a pair as non-overlapping if its two members invoke no heuristic in common. Each bar conditions on the number of heuristics a focal investor invokes. The bar labeled k reports the proportion of non-overlapping pairs among all pairs in which at least one member invokes exactly k heuristics, and “4+” denotes four or more. A pair can enter more than one bar.

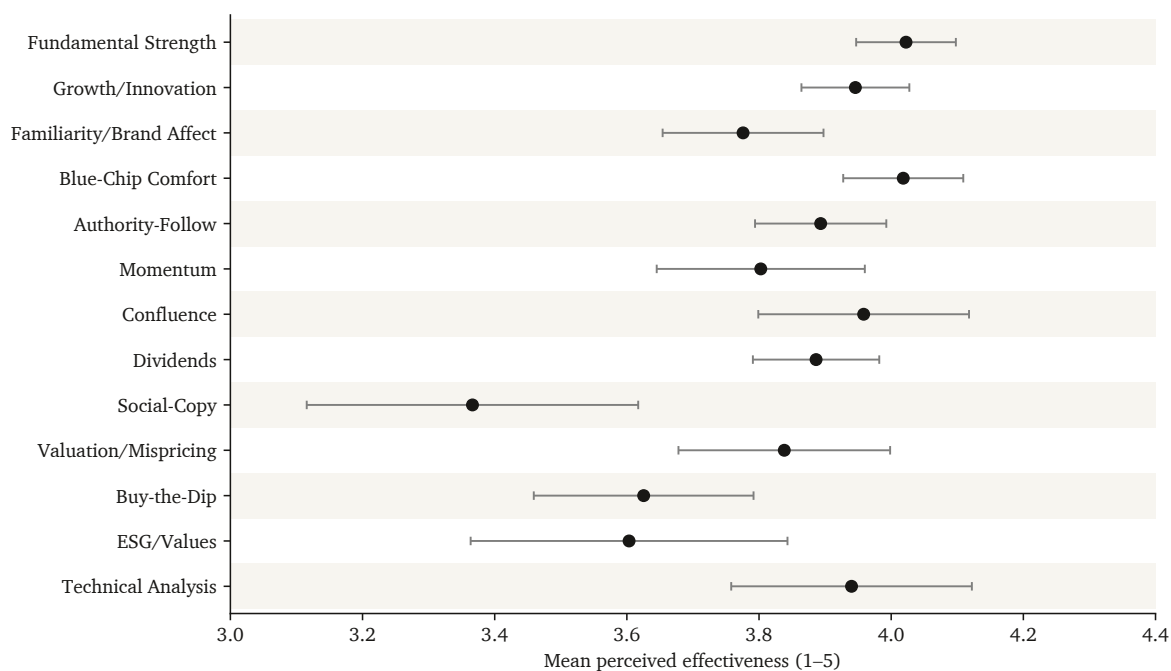


Figure F6. Perceived Heuristic Effectiveness

This figure shows how effective the follow-up survey respondents believe each of the thirteen heuristics to be. On a five-point scale from 1 (“not effective at all”) to 5 (“extremely effective”), respondents rate how effective a heuristic is at finding stocks that match an investment objective in their own local stock market. The rated heuristics are those we identify in the interview of the respondent plus any others the respondent reports having used in the past. When these exceed three, the respondent rates only the three longest-used. We elicit ratings only for the objectives the respondent rates 4 or 5 in importance, so every rating comes from a respondent who has used the heuristic and considers the objective important. Twenty-one respondents rate no objective that highly and therefore rate no heuristic. The figure pools the four objectives. Because a respondent rates between one and four objectives, we first average the ratings of each respondent for a heuristic across the objectives she rates and then average across respondents, so that each respondent counts equally. Markers report these means and horizontal bars the associated 95% confidence intervals, and we truncate the horizontal axis to the range of the estimates. All thirteen means lie above the midpoint of the scale. The number of respondents behind a heuristic ranges from 25 to 263. We order heuristics from top to bottom by their prevalence in the full interview sample.

Table F1. Interview Duration

This table examines whether the duration of the interview varies across investor demographics and countries. The dependent variable is the interview duration in minutes, the elapsed time between the first and the last message of the interview transcript, without winsorization. We estimate ordinary least squares regressions. Column (1) includes the six demographic indicators as independent variables, and Column (2) adds country indicators, with the US as the omitted category. The six demographic indicators are *Male*, *Older* (age ≥ 35), *Higher education* (bachelor's degree or above), *High wealth* (investable assets $\geq$ USD 500,000), *Advanced* (self-reported advanced or expert investment knowledge), and *Aggressive* (self-reported willingness to take above-average or substantial financial risks). We demean them so that the intercept in Column (1) represents the unconditional sample average. We do not demean the country indicators, so the intercept in Column (2) is the fitted duration for a US investor with sample-average demographics. The sample comprises the 1,540 investors we interviewed. We report heteroskedasticity-robust standard errors in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: Interview Duration (Minutes)		
	(1)	(2)
Male	+1.977*** (0.747)	+1.570** (0.772)
Older	+2.557*** (0.829)	+3.003*** (0.854)
Higher education	-1.049 (0.931)	-1.575 (0.968)
High wealth	+0.468 (0.786)	+0.700 (0.776)
Advanced	+0.314 (0.930)	-0.328 (1.011)
Aggressive	+0.429 (0.804)	+0.078 (0.787)
Australia		+1.004 (1.586)
Canada		-0.350 (1.420)
France		-1.138 (1.086)
Germany		+1.528 (1.458)
India		+5.374*** (1.432)
Japan		+4.853*** (1.433)
Singapore		-1.922 (1.545)
South Korea		+0.119 (1.215)
UK		+0.164 (1.493)
Intercept	+18.738*** (0.350)	+17.862*** (0.730)
Observations	1,540	1,540
R^2	0.014	0.037

Table F2. Preference for an AI versus a Human Interviewer

This table examines whether the preference for an AI over a human interviewer varies across investor demographics and countries. The dependent variable equals 100 if the investor answers that she would prefer to be interviewed by an AI interviewer in the future, and zero if she prefers a human interviewer, is indifferent, or gives no usable answer (168, 766, and 20 investors, respectively). We therefore drop no investor, and the coefficients are in percentage points. We estimate linear probability models. Column (1) includes the six demographic indicators as independent variables, and Column (2) adds country indicators, with the US as the omitted category. The six demographic indicators are *Male*, *Older* (age ≥ 35), *Higher education* (bachelor's degree or above), *High wealth* (investable assets $\geq$ USD 500,000), *Advanced* (self-reported advanced or expert investment knowledge), and *Aggressive* (self-reported willingness to take above-average or substantial financial risks). We demean them so that the intercept in Column (1) represents the unconditional sample average. We do not demean the country indicators, so the intercept in Column (2) is the fitted value for a US investor with sample-average demographics. The sample comprises the 1,540 investors we interviewed. We report heteroskedasticity-robust standard errors in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: Prefer AI Interviewer		
	(1)	(2)
Male	+3.813 (2.647)	+2.524 (2.655)
Older	-11.957*** (2.963)	-9.952*** (2.969)
Higher education	+8.475*** (3.018)	+5.359* (3.039)
High wealth	-1.259 (2.647)	-0.096 (2.591)
Advanced	+4.306 (3.187)	+1.837 (3.314)
Aggressive	+4.782* (2.720)	+2.777 (2.726)
Australia		-4.705 (4.699)
Canada		+4.720 (4.883)
France		-11.704*** (4.443)
Germany		+8.828* (5.078)
India		+28.832*** (5.349)
Japan		+11.794** (5.256)
Singapore		+9.509* (5.070)
South Korea		+18.263*** (5.171)
UK		+1.191 (4.823)
Intercept	+38.052*** (1.224)	+31.986*** (2.834)
Observations	1,540	1,540
R^2	0.025	0.070

Table F3. Follow-Up Survey Demographics

This table shows descriptive statistics for the 600 investors who completed the follow-up survey. The first row covers the full follow-up survey sample. The remaining rows group these investors by gender, age, education, investable assets, self-reported investment knowledge, and self-reported risk attitude, each of which we elicit in the background survey. We leave the percentage columns blank for groups with no investors. Columns (1) and (2) report the number of investors in each group and that number as a percentage of the sample. Columns (3) through (6) report the percentage of investors within each group who report owning individual stocks, mutual funds or exchange-traded funds, derivatives, and cryptocurrencies, respectively. Column (3) equals 100% by construction, because we filter out investors who do not own any individual stocks before the interview stage.

	N (1)	% Sample (2)	% Owning			
			Stocks (3)	Funds (4)	Deriv. (5)	Crypto (6)
Overall	600	100.0	100.0	66.5	14.8	35.5
Gender						
Female	208	34.7	100.0	64.9	11.5	23.6
Male	392	65.3	100.0	67.3	16.6	41.8
Other	0					
Age						
21–24	12	2.0	100.0	58.3	16.7	75.0
25–34	91	15.2	100.0	82.4	28.6	50.5
35–44	139	23.2	100.0	77.0	26.6	49.6
45–54	138	23.0	100.0	60.9	9.4	38.4
55–64	105	17.5	100.0	56.2	7.6	23.8
65+	115	19.2	100.0	58.3	2.6	9.6
Education						
No degree	11	1.8	100.0	54.5	27.3	18.2
School diploma	111	18.5	100.0	48.6	8.1	23.4
Bachelor's	290	48.3	100.0	65.5	12.1	39.3
Master's	157	26.2	100.0	79.0	22.3	36.9
Doctorate	22	3.7	100.0	90.9	27.3	45.5
Other	9	1.5	100.0	55.6	11.1	33.3
Investable assets						
1K–10K	36	6.0	100.0	41.7	16.7	36.1
10K–25K	33	5.5	100.0	45.5	9.1	24.2
25K–50K	43	7.2	100.0	58.1	9.3	37.2
50K–100K	42	7.0	100.0	50.0	7.1	40.5
100K–250K	41	6.8	100.0	63.4	17.1	43.9
250K–500K	28	4.7	100.0	60.7	21.4	39.3
500K–1M	180	30.0	100.0	66.7	15.0	40.0
1M+	197	32.8	100.0	81.2	16.8	29.4
Investment knowledge						
Complete beginner	17	2.8	100.0	41.2	0.0	17.6
Rudimentary	100	16.7	100.0	44.0	6.0	18.0
Intermediate	318	53.0	100.0	66.4	11.6	34.3
Advanced	143	23.8	100.0	83.2	25.2	49.7
Expert	22	3.7	100.0	81.8	45.5	54.5
Risk attitude						
Conservative	32	5.3	100.0	43.8	9.4	12.5
Average	307	51.2	100.0	58.0	7.5	24.8
Aggressive	222	37.0	100.0	77.9	22.1	49.1
Very aggressive	39	6.5	100.0	87.2	35.9	61.5

Table F4. Objective Importance by Country

This table shows, separately by country, the percentage of follow-up survey respondents who consider each of four investment objectives important. On a five-point scale from 1 (“not important at all”) to 5 (“extremely important”), respondents rate how important each objective is to them personally when picking stocks (consistent high returns, high upside focus, stable performance, and personal values alignment). The first row, *Overall*, pools all 600 respondents. Each of the remaining ten rows corresponds to one country of residence. Column (1) reports the number of respondents in the country, and Columns (2) through (5) report the percentage of those respondents assigning a rating of 4 or 5 to the objective in the column header.

	N (1)	Consistent High Returns (2)	High Upside Focus (3)	Stable Performance (4)	Personal Values Alignment (5)
Overall	600	71.2	63.0	77.0	40.2
Australia	21	61.9	42.9	66.7	33.3
Canada	86	70.9	62.8	76.7	32.6
France	63	69.8	68.3	76.2	44.4
Germany	69	65.2	59.4	71.0	42.0
India	66	87.9	80.3	83.3	68.2
Japan	12	75.0	58.3	66.7	66.7
Singapore	30	73.3	76.7	83.3	63.3
South Korea	21	85.7	76.2	81.0	42.9
UK	78	60.3	51.3	70.5	29.5
US	154	71.4	59.7	81.2	29.2

Table F5. Follow-Up Survey Participation

This table compares the 600 investors who participated in the follow-up survey (*Respondents*) with the 940 investors who did not (*Non-respondents*). In each panel, Columns (1) and (2) report the group means, and Column (3) reports the difference between respondents and non-respondents. Panel A covers characteristics we measure at the interview stage. Its first row reports the number of investors in each group, and the next six rows report the percentage of investors in each group who are male (*Male*), are aged 35 or above (*Age 35+*), hold a bachelor's degree or above (*Bachelor's or higher*), report investable assets of at least USD 500,000 (*High assets*), report advanced or expert investment knowledge (*Advanced knowledge*), and report willingness to take above-average or substantial financial risks (*Aggressive risk*). *Interview minutes* is the elapsed duration of the interview in minutes, and *Number of heuristics* is the number of the thirteen heuristics the investor invokes. An investor invokes a heuristic if its score is at least three ("meaningful contributor"). Panel B reports, for each of the thirteen heuristics, the percentage of investors in each group who invoke it, and we order heuristics by their prevalence in the full interview sample. We test each difference with a two-sided two-sample *t*-test that allows unequal variances. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Panel A: Investor Characteristics

	Follow-up survey		Difference (3)
	Respondents (1)	Non-respondents (2)	
Number of investors	600	940	
Male (%)	65.3	67.3	-2.0
Age 35+ (%)	82.8	69.9	12.9***
Bachelor's or higher (%)	78.2	80.1	-1.9
High assets (%)	62.8	53.5	9.3***
Advanced knowledge (%)	27.5	21.1	6.4***
Aggressive risk (%)	43.5	40.4	3.1
Interview minutes	18.5	18.9	-0.3
Number of heuristics	2.56	2.57	0.00

Panel B: Heuristic Adoption

	Follow-up survey		Difference (3)
	Respondents (1)	Non-respondents (2)	
Fundamental Strength	39.3	41.2	-1.8
Growth/Innovation	35.5	38.9	-3.4
Familiarity/Brand Affect	28.7	28.3	0.4
Blue-Chip Comfort	27.2	28.7	-1.6
Authority-Follow	20.7	17.3	3.3
Momentum	17.0	18.5	-1.5
Confluence	14.0	15.0	-1.0
Dividends	14.2	13.9	0.2
Social-Copy	14.7	13.0	1.7
Valuation/Mispricing	15.0	12.7	2.3
Buy-the-Dip	13.2	12.0	1.1
ESG/Values	11.8	9.4	2.5
Technical Analysis	5.2	7.8	-2.6**

Table F6. Heuristic Adoption: Logit Estimates

This table examines whether investors adopt the heuristics they perceive as effective for the objectives they consider important, re-estimating the linear probability models of Table 6 of the main paper as fixed-effects logit models. We estimate the following logit counterpart of equation (1) of the main paper:

$$\Pr(D(\text{Adoption})_{ih} = 1) = \Lambda\left(\beta_1 \left(\theta_{ig} \times \eta_{ihg}^p\right) + \beta_2 \eta_{ihg}^p + \beta_3 \theta_{ig} + \alpha_i\right),$$

in which $\Lambda(\cdot)$ denotes the logistic function, and the dependent variable, $D(\text{Adoption})_{ih}$, equals one if heuristic h is among those we identify in the interview of investor i . The key independent variable is the interaction $\theta_{ig} \times \eta_{ihg}^p$, printed as *Objective importance* $\times$ *Perceived effectiveness*, in which θ_{ig} is the five-point importance rating investor i gives to objective g , and η_{ihg}^p is the five-point rating she gives to how effective heuristic h is at finding stocks that match the objective in her own local stock market. We standardize both variables within investor, as in Table 6 of the main paper. The starting sample is that of Table 6 of the main paper, 3,744 investor–heuristic–objective observations from 572 investors. The logit estimation drops fixed-effect groups without variation in the outcome, because the group effect fits their observations perfectly, which leaves 3,291, 3,291, and 3,181 observations in Columns (1) through (3), respectively. Coefficients are in log-odds and are therefore not directly comparable in scale to the linear probability estimates. The row *Pseudo R*² reports McFadden’s pseudo-*R*², computed against an intercept-only logit on the estimation sample of each column. *Investor FE*, *Investor* $\times$ *Objective FE*, and *Country* $\times$ *Heuristic* $\times$ *Objective FE* refer to the corresponding fixed effects. We cluster standard errors at the investor level and report them in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: D(Adoption)			
	(1)	(2)	(3)
Objective importance $\times$ Perceived effectiveness	+0.174** (0.078)	+0.194** (0.088)	+0.264** (0.122)
Objective importance	-0.002 (0.003)		
Perceived effectiveness	-0.021 (0.060)	-0.027 (0.074)	-0.062 (0.100)
Investor FE	✓	–	–
Investor $\times$ Objective FE	–	✓	✓
Country $\times$ Heuristic $\times$ Objective FE	–	–	✓
Observations	3,291	3,291	3,181
Pseudo <i>R</i> ²	0.151	0.151	0.323

Table F7. Perceived and Historical Heuristic Effectiveness

This table examines the relationship between perceived and historical heuristic effectiveness. We estimate the following regression, equation (2) of the main paper:

$$\eta_{ihg}^p = \beta \eta_{hgc}^H + \alpha_i + \lambda_g + \varepsilon_{ihg},$$

in which the dependent variable η_{ihg}^p , *Perceived effectiveness*, is the five-point rating investor i gives to how effective heuristic h is at finding stocks that match objective g in her own local stock market, standardized within investor across all her ratings for the three financial objectives. The key independent variable η_{hgc}^H , *Historical effectiveness*, is the delivery of the heuristic on the objective in country c of the investor, estimated for the period running from July 2015 through June 2025, the ten years preceding our interviews (Section 5.3 of the main paper), and standardized within objective across the seventy heuristic–country pairs. The pooled sample comprises the 2,561 investor–heuristic–objective observations from 560 investors for which the follow-up survey elicits an effectiveness rating, restricted to the seven heuristics covered by archival data. Column (1) pools the three financial objectives and includes objective fixed effects λ_g , and Columns (2) through (4) estimate the relation separately for the returns, upside, and stability objectives. All columns include investor fixed effects α_i . We cluster standard errors at the country level and report them in parentheses. Given the small number of clusters (ten), we base inference on wild-cluster bootstrap p -values (Webb six-point weights, 9,999 replications), which are 0.001, 0.102, 0.026, and 0.012 in Columns (1) through (4). Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: Perceived effectiveness				
	(1) All	(2) Returns	(3) Upside	(4) Stability
Historical effectiveness	+0.137*** (0.016)	+0.128 (0.036)	+0.147** (0.040)	+0.209** (0.035)
Investor FE	✓	✓	✓	✓
Objective FE	✓	–	–	–
Observations	2,561	871	771	919
R^2	0.140	0.355	0.404	0.429

Table F8. Perceived and Historical Heuristic Effectiveness: Heterogeneity across Investor Groups

This table examines whether the relationship between perceived and historical heuristic effectiveness varies with investor characteristics. The dependent variable, *Perceived effectiveness*, and the key independent variable, *Historical effectiveness*, are as in Table F7. Column (1) reproduces Column (1) of Table F7, which pools the three financial objectives with investor and objective fixed effects. Columns (2) through (7) each add one interaction between historical effectiveness and an investor characteristic, and Column (8) includes all six interactions jointly. The six characteristics are the indicators *Male*, *Older* (age ≥ 35), *Higher education* (bachelor's degree or above), *High wealth* (investable assets $\geq$ USD 500,000), *Advanced knowledge* (self-reported advanced or expert investment knowledge), and *Aggressive* (self-reported willingness to take above-average or substantial financial risks). The investor fixed effects absorb their main effects. The sample comprises the 2,561 investor–heuristic–objective observations from 560 investors in Column (1) of Table F7. We cluster standard errors at the country level and report them in parentheses. Given the small number of clusters (ten), we base inference on wild-cluster bootstrap *p*-values (Webb six-point weights, 9,999 replications). Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: Perceived effectiveness								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Historical effectiveness	+0.137*** (0.016)	+0.158*** (0.024)	+0.150*** (0.022)	+0.190** (0.042)	+0.107** (0.037)	+0.138** (0.022)	+0.147** (0.022)	+0.204** (0.065)
Historical effectiveness $\times$ Male		-0.031 (0.026)						-0.028 (0.024)
Historical effectiveness $\times$ Older			-0.015 (0.030)					-0.025 (0.030)
Historical effectiveness $\times$ Higher education				-0.069 (0.055)				-0.080 (0.047)
Historical effectiveness $\times$ High wealth					+0.043 (0.046)			+0.062 (0.041)
Historical effectiveness $\times$ Advanced knowledge						-0.004 (0.046)		+0.009 (0.041)
Historical effectiveness $\times$ Aggressive							-0.024 (0.035)	-0.026 (0.031)
Investor FE	✓	✓	✓	✓	✓	✓	✓	✓
Objective FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	2,561	2,561	2,561	2,561	2,561	2,561	2,561	2,561
R ²	0.140	0.140	0.140	0.141	0.140	0.140	0.140	0.142

Table F9. Do Adopted Heuristics Deliver? Average Stock and JKP-Factor Benchmarks

This table reports two additional benchmarks for Table 7 of the main paper, using historical effectiveness over July 2015 through June 2025, the ten years preceding our interviews. The row *Adopted heuristics* repeats that of Table 7, with the same sample, prioritized objectives, adopted heuristics, aggregation, and standardization. The row *Average stock* reports the delivery of random stock sorting. Its long–short performance is zero in expectation before standardization, and we standardize it with the same within-objective means and standard deviations as heuristic effectiveness. This benchmark is a constant, so it has no standard deviation. Its one-sided p -value tests whether the adopted heuristics deliver more, using a wild-cluster bootstrap by country (Webb six-point weights, 9,999 replications). The row *Randomly selected JKP-factor-based portfolios* replaces the adopted set of each investor with an equally sized random draw from the 153 factors of Jensen, Kelly, and Pedersen (2023) that are computable in her country, each in its published long–short direction and standardized with the same within-objective means and standard deviations as heuristic effectiveness. This row reports the mean across 10,000 draws, with the standard deviation across draws in parentheses, and its p -value is the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises the 427 follow-up respondents of Table 7, and the columns *Returns*, *Upside*, and *Stability* use 210, 158, and 256 of them, respectively. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	+0.130	+0.009	+0.158	+0.179
Average stock	-0.262	-0.617	+0.842	-0.649
$p(\text{Adopted} \leq \text{average stock})$	[0.002]***	[0.006]***	[1.000]	[0.001]***
Randomly selected JKP-factor-based portfolios	-0.085	-0.233	+0.552	-0.354
Standard deviation across draws	(0.034)	(0.058)	(0.063)	(0.047)
$p(\text{Adopted} \leq \text{randomly selected JKP factors})$	[0.000]***	[0.000]***	[1.000]	[0.000]***

Table F10. Do Adopted Heuristics Deliver? Average Stock and JKP-Factor Benchmarks: Out-of-Sample Evidence

This table reports two additional benchmarks for Table 8 of the main paper, using out-of-sample delivery. The out-of-sample period covers portfolio sorts from September through November 2025, and we measure delivery over the month following each sort, so that all outcomes fall between October and December 2025. We calculate the CAPM alpha for the returns objective with the market beta fixed at its estimate from September 2015 through August 2025. The row *Adopted heuristics* repeats that of Table 8, with the same sample, prioritized objectives, adopted heuristics, aggregation, and standardization. The row *Average stock* reports the delivery of random stock sorting. Its long–short performance is zero in expectation before standardization, and we standardize it with the same within-objective means and standard deviations as heuristic effectiveness. This benchmark is a constant, so it has no standard deviation. Its one-sided p -value tests whether the adopted heuristics deliver more, using a wild-cluster bootstrap by country (Webb six-point weights, 9,999 replications). The row *Randomly selected JKP-factor-based portfolios* replaces the adopted set of each investor with an equally sized random draw from the 153 factors of Jensen, Kelly, and Pedersen (2023) that are computable in her country, each in its published long–short direction and standardized with the same within-objective means and standard deviations as heuristic effectiveness. This row reports the mean across 10,000 draws, with the standard deviation across draws in parentheses, and its p -value is the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises the 427 follow-up respondents of Table 8, and the columns *Returns*, *Upside*, and *Stability* use 210, 158, and 256 of them, respectively. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	+0.132	+0.058	+0.288	+0.099
Average stock $p(\text{Adopted} \leq \text{average stock})$	-0.163 [0.001]***	-0.263 [0.007]***	+0.699 [0.989]	-0.600 [0.001]***
Randomly selected JKP-factor-based portfolios	-0.060	-0.079	+0.546	-0.406
Standard deviation across draws	(0.033)	(0.059)	(0.064)	(0.045)
$p(\text{Adopted} \leq \text{randomly selected JKP factors})$	[0.000]***	[0.011]**	[1.000]	[0.000]***

Table F11. Do Adopted Heuristics Deliver? Post-Adoption Evidence

This table repeats the analysis of Table 8 of the main paper over post-adoption periods based on reported use durations. For each adopted heuristic, we use the duration of current use or, if the investor has stopped using it, the duration before stopping. We exclude investor–heuristic pairs that report a few months or an unknown duration. Taking June 30, 2026 as the common reference date, we use minimum durations of one and three years for reports of one to three years and more than three years, respectively. Monthly portfolio sorts therefore begin in July 2025 or July 2023, and we measure delivery over the month following each sort. Outcomes cover August through December 2025 (five months) or August 2023 through December 2025 (29 months, the longest period). The average period is approximately 20 months per investor, with retained heuristics weighted equally within each investor. We evaluate the choices recorded in the interviews through December 2025, including heuristics no longer used. For returns, we calculate the CAPM alpha with market betas fixed at their heuristic–country estimates from July 2013 through June 2023 portfolio sorts. For upside and stability, we use the top-1% hit-rate spread and the sign-reversed total return volatility spread. We standardize outcomes in both periods using the within-objective means and sample standard deviations across the seventy heuristic–country pairs in the 29-month window. Prioritized objectives and aggregation are otherwise as in Table 8 of the main paper. The two benchmark rows replace the retained adopted set of each investor with an equally sized random draw without replacement from (i) the covered heuristics she did not adopt in the interview (*Alternative heuristics*) and (ii) all seven covered heuristics (*Randomly selected heuristics*). The first pool also excludes adopted heuristics dropped by the duration filter. We randomly assign the drawn heuristics to the periods of the adopted heuristics and evaluate them in the same country on the same prioritized objectives. These two rows report means across 10,000 draws, with standard deviations across draws in parentheses, and their p -values are the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises 371 investors and 621 retained investor–heuristic pairs. The columns *Returns*, *Upside*, and *Stability* use 180, 137, and 224 of these investors, respectively. Interpreting these results as post-adoption delivery assumes that the reported durations identify the start of use and that the reported priorities apply during the evaluation period. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	+0.146	+0.084	+0.255	+0.117
Alternative heuristics	-0.122	-0.224	-0.172	+0.012
Standard deviation across draws	(0.041)	(0.081)	(0.080)	(0.051)
$p(\text{Adopted} \leq \text{alternative heuristics})$	[0.000]***	[0.000]***	[0.000]***	[0.021]**
Randomly selected heuristics	-0.066	-0.151	-0.086	+0.027
Standard deviation across draws	(0.044)	(0.085)	(0.086)	(0.056)
$p(\text{Adopted} \leq \text{randomly selected heuristics})$	[0.000]***	[0.003]***	[0.000]***	[0.051]*

Table F12. Determinants of Objective Priorities

This table examines which financial objectives investors prioritize. We estimate linear probability models for each objective. The dependent variable, $D(\text{investor prioritizes the objective})$, is a dummy variable equal to one if the investor rates the objective strictly above her own average importance rating across the three financial objectives. The six demographic indicators are *Male*, *Older* (age ≥ 35), *Higher education* (bachelor's degree or above), *High wealth* (investable assets $\geq$ USD 500,000), *Advanced knowledge* (self-reported advanced or expert investment knowledge), and *Aggressive* (self-reported willingness to take above-average or substantial financial risks). *Mkt excess return*, *Avg skewness*, and *Avg volatility* are country-level averages over July 2015 through June 2025, the ten years preceding our interviews, of the annualized value-weighted market excess return and the value-weighted means of the 21-day realized skewness and volatility of individual stocks. We compute all three from the data of Jensen, Kelly, and Pedersen (2023) and standardize them across the ten countries. For each objective, the first column includes the three market characteristics, and the second replaces them with country indicators, with the US as the omitted category. All six columns include the six demographic indicators and a constant. We report heteroskedasticity-robust standard errors in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: $D(\text{investor prioritizes the objective})$						
	Returns		Upside		Stability	
	(1)	(2)	(3)	(4)	(5)	(6)
Male	-0.080*	-0.077*	+0.077**	+0.077**	-0.071	-0.070
	(0.043)	(0.044)	(0.038)	(0.039)	(0.044)	(0.044)
Older	+0.005	-0.002	+0.015	+0.022	+0.031	+0.041
	(0.054)	(0.055)	(0.050)	(0.051)	(0.056)	(0.058)
Higher education	-0.006	-0.005	+0.111***	+0.109**	-0.078	-0.074
	(0.052)	(0.053)	(0.043)	(0.044)	(0.053)	(0.054)
High wealth	-0.053	-0.048	-0.011	-0.008	+0.045	+0.046
	(0.045)	(0.045)	(0.040)	(0.041)	(0.045)	(0.045)
Advanced knowledge	+0.075	+0.078	+0.053	+0.050	-0.022	-0.038
	(0.051)	(0.054)	(0.048)	(0.052)	(0.051)	(0.053)
Aggressive	+0.021	+0.018	+0.103**	+0.102**	-0.109**	-0.109**
	(0.045)	(0.046)	(0.042)	(0.043)	(0.046)	(0.047)
Mkt excess return	-0.018		-0.010		-0.041	
	(0.026)		(0.024)		(0.026)	
Avg skewness	-0.031		+0.002		+0.002	
	(0.031)		(0.029)		(0.032)	
Avg volatility	+0.029		-0.010		+0.021	
	(0.042)		(0.038)		(0.043)	
Australia		-0.094		-0.057		-0.048
		(0.108)		(0.098)		(0.120)
Canada		+0.013		+0.066		+0.051
		(0.068)		(0.061)		(0.068)
France		-0.053		+0.044		+0.004
		(0.073)		(0.071)		(0.075)
Germany		+0.008		+0.013		+0.083
		(0.076)		(0.068)		(0.077)
India		-0.092		+0.027		+0.056
		(0.081)		(0.077)		(0.083)
Japan		+0.073		+0.008		+0.104
		(0.155)		(0.142)		(0.147)
Singapore		-0.043		+0.024		+0.075
		(0.099)		(0.089)		(0.099)
South Korea		+0.022		-0.004		+0.014
		(0.117)		(0.103)		(0.117)
UK		+0.014		+0.055		+0.156**
		(0.070)		(0.062)		(0.068)
Constant	+0.449***	+0.464***	+0.094	+0.062	+0.614***	+0.544***
	(0.071)	(0.081)	(0.061)	(0.069)	(0.071)	(0.082)
Observations	600	600	600	600	600	600
R^2	0.012	0.016	0.045	0.048	0.031	0.038

Table F13. Heuristic Adoption and Historical Effectiveness: Reduced Form

This table examines whether investors are more likely to adopt heuristics that are historically more effective, in their own country, for the objectives they consider important. It re-estimates the adoption regression of Table 6 of the main paper in reduced form, replacing perceived effectiveness η_{ihg}^P with historical effectiveness η_{hgc}^H . Because we estimate historical effectiveness from archival data alone (Section 5.3 of the main paper), survey answers cannot affect it, and the estimates are immune to ex-post rationalization of one's own choices. The dependent variable, $D(\text{Adoption})$, is a dummy variable equal to one if the heuristic is among those we identify in the interview of the investor. The key independent variable is the interaction *Objective importance* $\times$ *Historical effectiveness*, in which *Objective importance* is the five-point importance rating the investor gives to the objective, standardized within investor across the three financial objectives, and *Historical effectiveness* is the delivery of the heuristic on the objective in the country of the investor, estimated for the period running from July 2015 through June 2025, the ten years preceding our interviews, and standardized within objective across the seventy heuristic–country pairs. Because the analysis requires no effectiveness rating, the sample is the full grid of the 600 follow-up respondents, the seven heuristics covered by archival data, and the three financial objectives (12,600 observations), without any conditioning on which heuristics an investor rates. *Investor FE*, *Investor* $\times$ *Objective FE*, and *Country* $\times$ *Heuristic* $\times$ *Objective FE* refer to the corresponding fixed effects. We cluster standard errors at the country level and report them in parentheses. Given the small number of clusters (ten), we base the stars in Columns (1) and (2) on wild-cluster bootstrap p -values (Webb six-point weights, 9,999 replications). In Column (3), the bootstrap is not computable under the high-dimensional fixed effects, so we base the stars on the country-clustered standard errors. The interaction remains significant at the 1% level when we instead cluster standard errors at the investor level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: D(Adoption)			
	(1)	(2)	(3)
Objective importance $\times$ Historical effectiveness	+0.024*** (0.005)	+0.025*** (0.005)	+0.025*** (0.005)
Historical effectiveness	+0.004 (0.007)	+0.005 (0.008)	
Objective importance	+0.001 (0.001)		
Investor FE	✓	–	–
Investor $\times$ Objective FE	–	✓	✓
Country $\times$ Heuristic $\times$ Objective FE	–	–	✓
Observations	12,600	12,600	12,600
R^2	0.099	0.099	0.173

Table F14. Heuristic Adoption and Historical Effectiveness: Reduced Form by Objective

This table repeats the reduced-form adoption analysis of Table F13 separately for each financial objective. In each column, we regress $D(\text{Adoption})$ on the interaction between the importance rating of the objective, standardized within investor across the three financial objectives, and the historical effectiveness of the heuristic for that objective in the country of the investor. Each column uses the full grid of the 600 follow-up respondents and the seven covered heuristics for the corresponding objective (4,200 observations). *Investor FE* and *Country $\times$ Heuristic FE* refer to the corresponding fixed effects. We cluster standard errors at the country level and report them in parentheses. The wild-cluster bootstrap is not computable under these fixed effects, so we base the stars on the country-clustered standard errors. The interactions for upside and stability remain significant at the 10% level when we instead cluster standard errors at the investor level. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Dependent variable: $D(\text{Adoption})$			
	(1) Returns	(2) Upside	(3) Stability
Objective importance $\times$ Historical effectiveness	+0.014*** (0.004)	+0.032*** (0.007)	+0.027*** (0.005)
Investor FE	✓	✓	✓
Country $\times$ Heuristic FE	✓	✓	✓
Observations	4,200	4,200	4,200
R^2	0.172	0.175	0.174

Table F15. Do Adopted Heuristics Deliver on Non-Prioritized Objectives?

This table repeats the analysis of Table 7 of the main paper on the objectives investors do *not* prioritize. An objective counts as non-prioritized if the investor rates it strictly below her own average across the three financial objectives. The adopted heuristics of each investor are those we identify in her interview among the seven covered by archival data (Table 5 of the main paper). The row *Adopted heuristics* reports the delivery of these heuristics on the non-prioritized objectives in the country of the investor. It averages over investors, in the column *Overall* after first averaging the non-prioritized objectives of each investor so that each investor enters once, and in the other columns over the investors for whom that objective is non-prioritized. The design is otherwise identical to Table 7 of the main paper. The two benchmark rows replace the adopted rule set of each investor with an equally sized random draw from (i) the covered heuristics she does not adopt (*Alternative heuristics*) and (ii) all seven covered heuristics (*Randomly selected heuristics*). These two rows report means across 10,000 draws, with standard deviations across draws in parentheses, and their *p*-values are the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises the 427 follow-up respondents who adopt at least one covered heuristic and have at least one non-prioritized financial objective, excluding adopters of four or more of the seven. The columns *Returns*, *Upside*, and *Stability* use 184, 255, and 146 of them, respectively. Table F16 reports the average-stock and JKP-factor-based portfolio benchmarks, with the same adopted-heuristics row. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the *p*-values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	-0.100	-0.078	-0.127	-0.091
Alternative heuristics	-0.026	-0.124	+0.016	+0.017
Standard deviation across draws	(0.029)	(0.055)	(0.044)	(0.061)
<i>p</i> (Adopted $\leq$ alternative heuristics)	[0.994]	[0.195]	[0.999]	[0.962]
Randomly selected heuristics	-0.050	-0.140	-0.014	-0.007
Standard deviation across draws	(0.032)	(0.057)	(0.047)	(0.063)
<i>p</i> (Adopted $\leq$ randomly selected heuristics)	[0.942]	[0.135]	[0.991]	[0.906]

Table F16. Do Adopted Heuristics Deliver on Non-Prioritized Objectives? Average Stock and JKP-Factor Benchmarks

This table reports two additional benchmarks for Table F15, using historical effectiveness over July 2015 through June 2025, the ten years preceding our interviews. An objective counts as non-prioritized if the investor rates it strictly below her own average across the three financial objectives. The row *Adopted heuristics* repeats that of Table F15, with the same sample, non-prioritized objectives, adopted heuristics, aggregation, and standardization. The row *Average stock* reports the delivery of random stock sorting. Its long–short performance is zero in expectation before standardization, and we standardize it with the same within-objective means and standard deviations as heuristic effectiveness. This benchmark is a constant, so it has no standard deviation. Its one-sided p -value tests whether the adopted heuristics deliver more, using a wild-cluster bootstrap by country (Webb six-point weights, 9,999 replications). The row *Randomly selected JKP-factor-based portfolios* replaces the adopted set of each investor with an equally sized random draw from the 153 factors of Jensen, Kelly, and Pedersen (2023) that are computable in her country, each in its published long–short direction and standardized with the same within-objective means and standard deviations as heuristic effectiveness. This row reports the mean across 10,000 draws, with the standard deviation across draws in parentheses, and its p -value is the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises the 427 follow-up respondents of Table F15, and the columns *Returns*, *Upside*, and *Stability* use 184, 255, and 146 of them, respectively. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	-0.100	-0.078	-0.127	-0.091
Average stock	+0.043	-0.617	+0.842	-0.649
$p(\text{Adopted} \leq \text{average stock})$	[0.967]	[0.015]**	[1.000]	[0.000]***
Randomly selected JKP-factor-based portfolios	+0.084	-0.235	+0.526	-0.355
Standard deviation across draws	(0.035)	(0.064)	(0.052)	(0.063)
$p(\text{Adopted} \leq \text{randomly selected JKP factors})$	[1.000]	[0.007]***	[1.000]	[0.000]***

Table F17. Do Adopted Heuristics Deliver on Non-Prioritized Objectives? Out-of-Sample Evidence

This table repeats the out-of-sample analysis of Table 8 of the main paper on the objectives investors do *not* prioritize. We recompute delivery on outcomes realized *after* our interviews. The out-of-sample period covers portfolio sorts from September through November 2025, and we measure delivery over the month following each sort, so that all outcomes fall between October and December 2025. Because that period is too short to estimate a market beta, we calculate the CAPM alpha for the returns objective with the market beta fixed at its estimate from the preceding ten years (September 2015 through August 2025). We standardize each outcome within objective across the seventy heuristic–country pairs, recomputed on the out-of-sample window. An objective counts as non-prioritized if the investor rates it strictly below her own average across the three financial objectives. The adopted heuristics, the aggregation, and the samples are identical to Table F15. The two benchmark rows replace the adopted rule set of each investor with an equally sized random draw from (i) the covered heuristics she does not adopt (*Alternative heuristics*) and (ii) all seven covered heuristics (*Randomly selected heuristics*). These two rows report means across 10,000 draws, with standard deviations across draws in parentheses, and their p -values are the one-sided fraction of draws that deliver at least as much as the adopted set. Table F18 reports the average-stock and JKP-factor-based portfolio benchmarks, with the same adopted-heuristics row. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	-0.033	-0.082	+0.062	-0.137
Alternative heuristics	+0.015	-0.030	+0.050	+0.046
Standard deviation across draws	(0.028)	(0.047)	(0.043)	(0.059)
$p(\text{Adopted} \leq \text{alternative heuristics})$	[0.955]	[0.864]	[0.390]	[0.999]
Randomly selected heuristics	+0.008	-0.044	+0.055	+0.009
Standard deviation across draws	(0.030)	(0.048)	(0.046)	(0.062)
$p(\text{Adopted} \leq \text{randomly selected heuristics})$	[0.919]	[0.777]	[0.446]	[0.991]

Table F18. Do Adopted Heuristics Deliver on Non-Prioritized Objectives? Average Stock and JKP-Factor Benchmarks: Out-of-Sample Evidence

This table reports two additional benchmarks for Table F17, using out-of-sample delivery. The out-of-sample period covers portfolio sorts from September through November 2025, and we measure delivery over the month following each sort, so that all outcomes fall between October and December 2025. We calculate the CAPM alpha for the returns objective with the market beta fixed at its estimate from September 2015 through August 2025. An objective counts as non-prioritized if the investor rates it strictly below her own average across the three financial objectives. The row *Adopted heuristics* repeats that of Table F17, with the same sample, non-prioritized objectives, adopted heuristics, aggregation, and standardization. The row *Average stock* reports the delivery of random stock sorting. Its long-short performance is zero in expectation before standardization, and we standardize it with the same within-objective means and standard deviations as heuristic effectiveness. This benchmark is a constant, so it has no standard deviation. Its one-sided p -value tests whether the adopted heuristics deliver more, using a wild-cluster bootstrap by country (Webb six-point weights, 9,999 replications). The row *Randomly selected JKP-factor-based portfolios* replaces the adopted set of each investor with an equally sized random draw from the 153 factors of Jensen, Kelly, and Pedersen (2023) that are computable in her country, each in its published long-short direction and standardized with the same within-objective means and standard deviations as heuristic effectiveness. This row reports the mean across 10,000 draws, with the standard deviation across draws in parentheses, and its p -value is the one-sided fraction of draws that deliver at least as much as the adopted set. The sample comprises the 427 follow-up respondents of Table F17, and the columns *Returns*, *Upside*, and *Stability* use 184, 255, and 146 of them, respectively. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively, referring to the p -values in the brackets.

Delivery, in standard deviations from the average heuristic-country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
Adopted heuristics	-0.033	-0.082	+0.062	-0.137
Average stock $p(\text{Adopted} \leq \text{average stock})$	+0.096 [0.969]	-0.263 [0.141]	+0.699 [0.999]	-0.600 [0.002]***
Randomly selected JKP-factor-based portfolios Standard deviation across draws $p(\text{Adopted} \leq \text{randomly selected JKP factors})$	+0.119 (0.034) [1.000]	-0.084 (0.061) [0.494]	+0.532 (0.052) [1.000]	-0.402 (0.060) [0.000]***

Table F19. Do Adopted Heuristics Deliver? Awareness-Restricted Alternatives

This table repeats the comparisons of Tables 7 and 8 of the main paper with the alternative-heuristics benchmark restricted to approaches each investor reports being aware of in the follow-up survey. The adopted set consists of the investor’s interview-recorded heuristics among the seven covered by archival data. The row “Alternative heuristics” replaces this set with an equally sized set drawn randomly, without replacement, from covered heuristics not adopted in the interview that the investor reports having heard of in the follow-up survey. Previously used alternatives remain eligible. We evaluate both sets in the investor’s country on her prioritized objectives, defined as objectives rated strictly above her own average importance across returns, upside, and stability. The “Overall” column first averages performance across each investor’s prioritized objectives and then across investors; the other columns average across investors who prioritize the corresponding objective. Panel A uses historical performance from July 2015 through June 2025. Panel B uses performance from October through December 2025, with CAPM betas estimated over September 2015 through August 2025 and held fixed. Returns use CAPM alpha, upside uses the top-1% hit-rate spread, and stability uses the sign-reversed total return-volatility spread. We standardize performance within objective across the seventy heuristic–country pairs, separately for each panel. Each benchmark reports the mean across 10,000 draws, with the standard deviation across draws in parentheses. Bracketed one-sided p -values give the fractions of draws that perform at least as well as the adopted set. Both panels use the same 360 investors, retaining adopters of one to three covered heuristics who prioritize at least one financial objective and have enough known alternatives for an equally sized draw. The returns, upside, and stability columns include 178, 129, and 221 investors, respectively; these subsamples overlap. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively. p -values are displayed to three decimals; stars use unrounded values.

Delivery, in standard deviations from the average heuristic–country pair				
	Overall (1)	Returns (2)	Upside (3)	Stability (4)
<i>Panel A: Historical Performance</i>				
Adopted heuristics	+0.133	-0.023	+0.168	+0.200
Alternative heuristics	-0.043	-0.156	+0.008	+0.028
Standard deviation across draws	(0.030)	(0.044)	(0.052)	(0.046)
$p(\text{Adopted} \leq \text{alternative heuristics})$	[0.000]***	[0.001]***	[0.001]***	[0.000]***
<i>Panel B: Out-of-Sample Performance</i>				
Adopted heuristics	+0.161	+0.063	+0.341	+0.132
Alternative heuristics	-0.001	-0.123	+0.061	+0.063
Standard deviation across draws	(0.026)	(0.039)	(0.052)	(0.041)
$p(\text{Adopted} \leq \text{alternative heuristics})$	[0.000]***	[0.000]***	[0.000]***	[0.051]*

Online Appendix References

- Argyle, Lisa P., Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate, 2023, Out of one, many: Using language models to simulate human samples, *Political Analysis* 31, 337–351.
- Barberis, Nicholas, 2018, Psychology-Based Models of Asset Prices and Trading Volume, in B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, ed.: *Handbook of Behavioral Economics: Applications and Foundations 1*, vol. 1 of *Handbook of Behavioral Economics - Foundations and Applications 1* . pp. 79–175 (North-Holland).
- Brown, Tom, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al., 2020, Language models are few-shot learners, *Advances in Neural Information Processing Systems* 33, 1877–1901.
- Campbell, John Y., and Luis M. Viceira, 1999, Consumption and Portfolio Decisions when Expected Returns are Time Varying, *The Quarterly Journal of Economics* 114, 433–495.
- Google, 2024, Prompting Guide 101, https://services.google.com/fh/files/misc/gemini_for_workspace_prompt_guide_october_2024_digital_final.pdf.
- Han, Bing, David Hirshleifer, and Johan Walden, 2022, Social Transmission Bias and Investor Behavior, *Journal of Financial and Quantitative Analysis* 57, 390–412.
- Hofstede, G., G.J. Hofstede, and M. Minkov, 2010, *Cultures and Organizations: Software of the Mind, Third Edition* (McGraw Hill LLC).
- Jensen, Theis Ingerslev, Bryan Kelly, and Lasse Heje Pedersen, 2023, Is There a Replication Crisis in Finance?, *The Journal of Finance* 78, 2465–2518.
- Kong, Aobo, Shiwan Zhao, Hao Chen, Qicheng Li, Yong Qin, Ruiqi Sun, Xin Zhou, Enzhi Wang, and Xiaohang Dong, 2023, Better zero-shot reasoning with role-play prompting, *arXiv preprint arXiv:2308.07702*.
- MacCallum, Noah, and Julian Lee, 2025, GPT-4.1 Prompting Guide, https://cookbook.openai.com/examples/gpt4-1_prompting_guide.
- Minkov, Michael, and Anneli Kaasa, 2022, Do dimensions of culture exist objectively? a validation of the revised Minkov-Hofstede model of culture with World Values Survey items and scores for 102 countries, *Journal of International Management* 28, 100971.
- Shanahan, Murray, Kyle McDonell, and Laria Reynolds, 2023, Role play with large language models, *Nature* 623, 493–498.
- Wang, Xuezhi, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou, 2022, Self-consistency improves chain of thought reasoning in language models, *arXiv preprint arXiv:2203.11171*.
- Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al., 2022, Chain-of-thought prompting elicits reasoning in large language models, *Advances in Neural Information Processing Systems* 35, 24824–24837.
- Yao, Shunyu, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan, 2023, Tree of thoughts: Deliberate problem solving with large language models, *Advances in Neural Information Processing Systems* 36, 11809–11822.